



Time Series forecasting models

Advanced Machine Learning



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 - LSTM
 - LLMs



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Definition

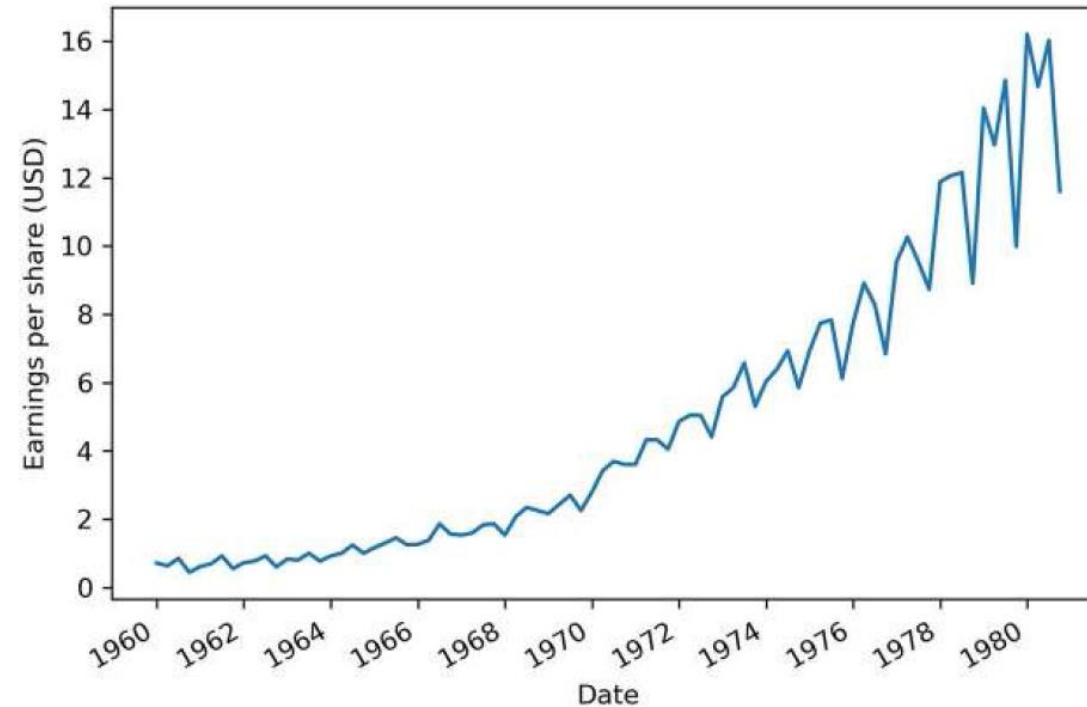


What is a time series?

- A set of data points **ordered** in **time**.
 - The data is often **equally spaced** in **time** → equal interval separate each data point.
 - The data can be recorded at **every hour** or every **minute**, or it could be **averaged** over every month or year.
- Common **applications**:
 - Closing value of a particular stock.
 - A household's electricity consumption.
 - Temperature outside.



Use case: The quarterly earnings per share of Johnson & Johnson



Quarterly earnings of Johnson & Johnson in USD from 1960 to 1980 showing a positive trend and a cyclical behavior.



Use case: The quarterly earnings per share of Johnson & Johnson (II)

- Data is **indexed** by **time**, as marked on the horizontal axis.
- Data is **equally spaced** in time → It was recorded at the end of **every quarter** of each **year**.
- There is a **trend** → The values are **increasing** over time.
- The earnings going up and down over the course of each year, and the **pattern** repeats every year.

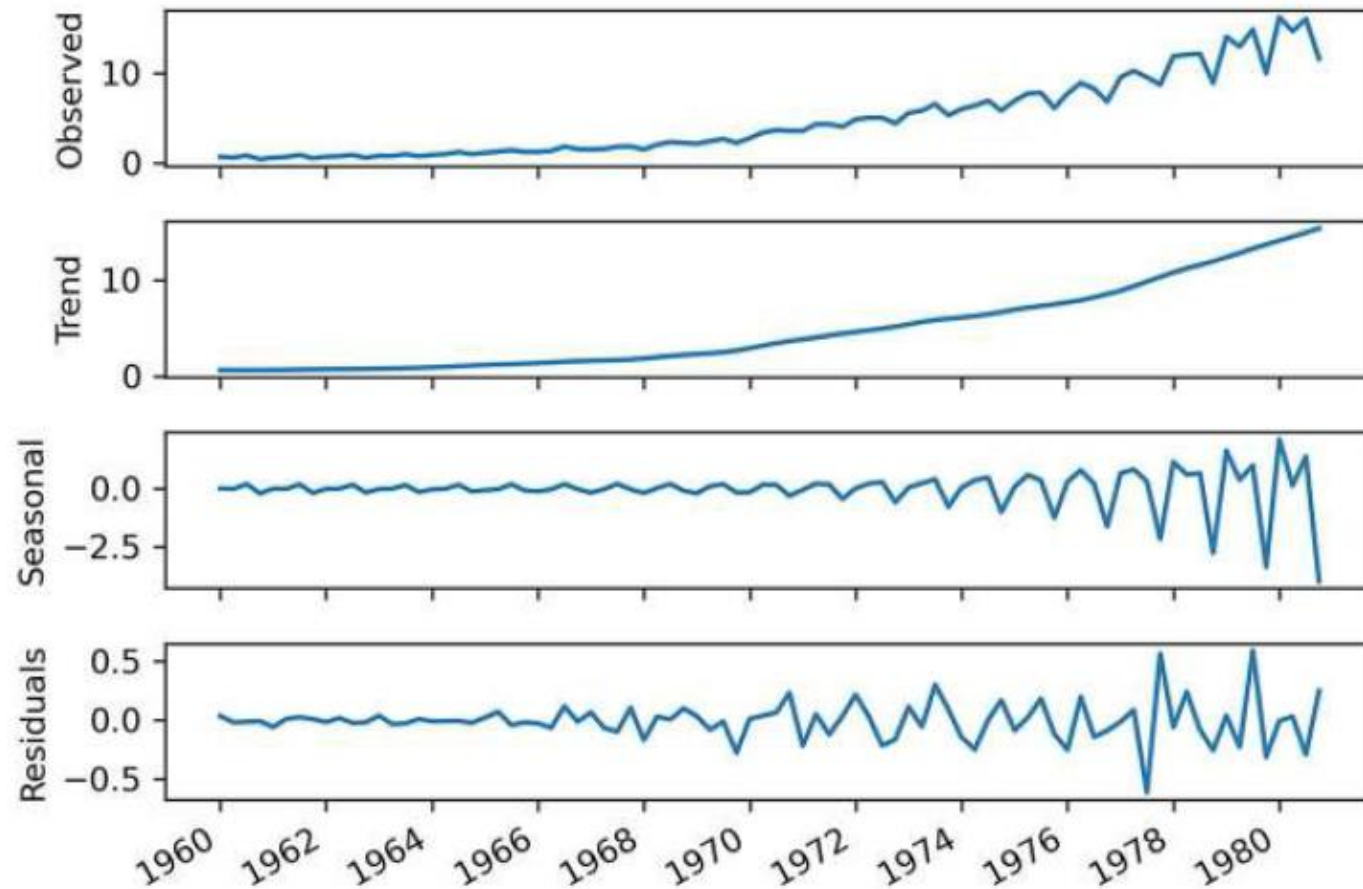


Components of a Time Series

- **Three main components** on which all time series can be decomposed:
 - **Trend.**
 - **Seasonal Component.**
 - **Residuals.**
- **Decomposition:** a statistical task that separates a time series into its different components.
 - We can visualize each individual component, which will help us identify the trend and seasonal pattern in the data → Not always straightforward.



Components of a Time Series (II)

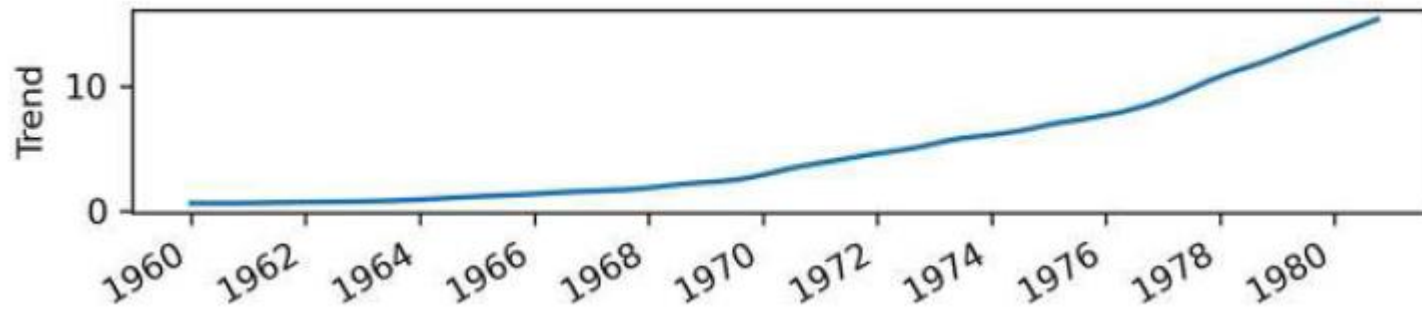


Decomposition of quarterly earnings of Johnson & Johnson in USD from 1960 to 1980.



Trend component

- It starts out flat and then steeply goes up → an increasing or **positive trend**.
- Trend component can be defined as **trying to draw a line through most of the data points** to show the **general direction** of a time series.

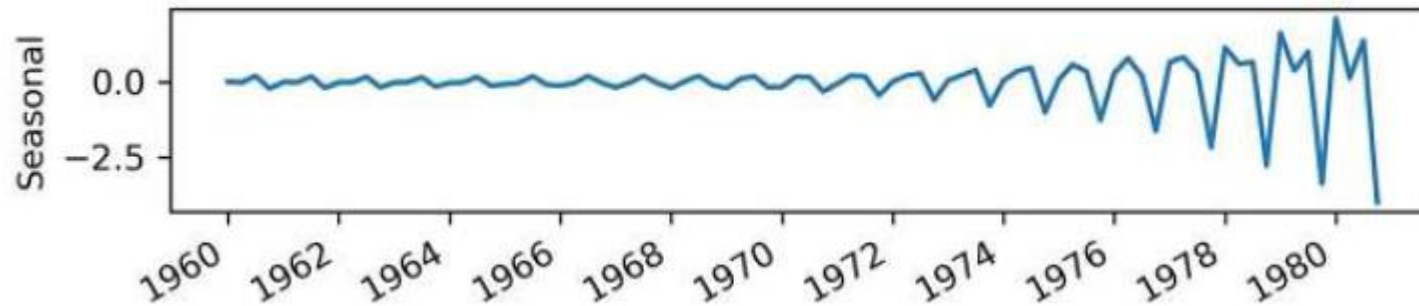


Trend component of quarterly earnings of Johnson & Johnson in USD from 1960 to 1980.



Seasonal component

- Captures the **seasonal** variation, which is a **cycle that occurs over a fixed period of time**.
 - For instance, we can detect seasons which only occur every four years → World Cup.
- We can see that over the course of a year, or four quarters, the earnings per share **start low, increase, and decrease** again at the end of the year.

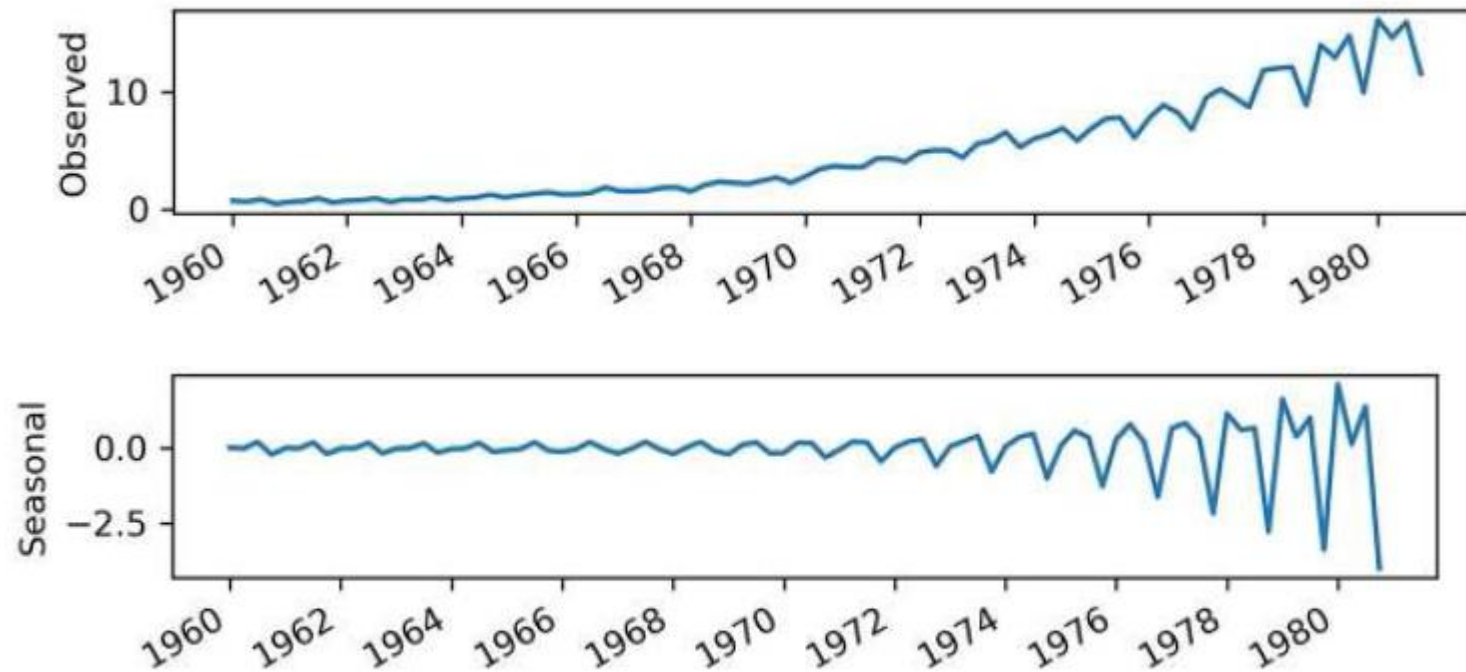


Seasonal component of quarterly earnings of Johnson & Johnson in USD from 1960 to 1980.



Seasonal component (II)

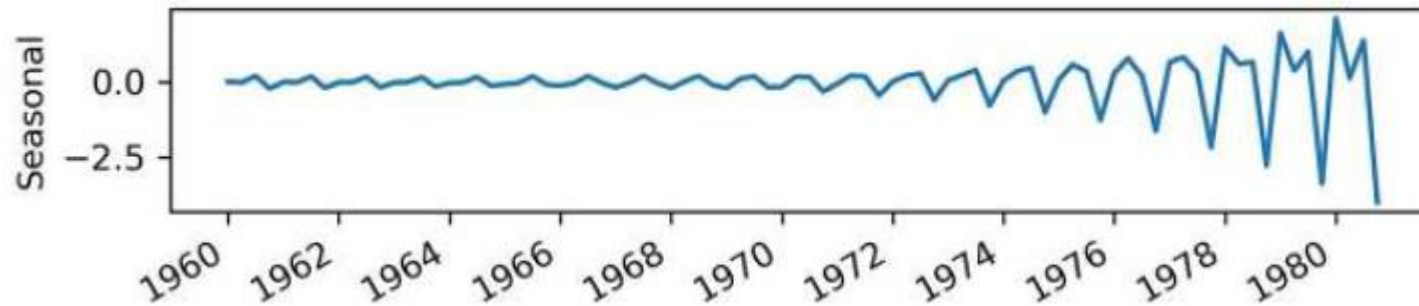
Which is the main difference between these two figures?





Seasonal component (III)

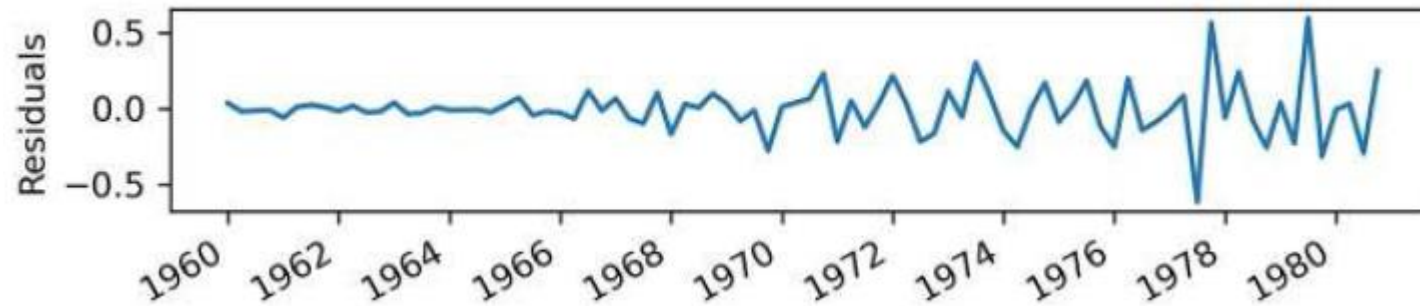
- The negative values mean → there is a **negative deviation from the trend**.
 - Sometimes we have a positive deviation, and we get a peak in the Observed graph. Other times, we have a negative deviation, and we see a trough in Observed.





Residual component

- Residuals usually correspond to **random errors**, also named as ***white noise***.
- It represents **information** that we **cannot model** or predict, since it is completely **random**.
- Residuals graph shows what value must be added to Trend and Seasonal in order to adjust the result and get the same value as in Observed.
 - Trend +





Decomposition methods

What are the methods to decompose the time series:

- Basic method
- STL (Seasonal + Trend decomp using LOESS)



Decomposition methods

Intuition behind basic method

- **Trend** → **Centered Moving Average (CMA)**. Moving window k for each value in x axes

$$T_t = \frac{1}{m} \sum_{j=-k}^k Y_{t+j}$$

- **Seasonality** → de-trend + average/cycle + normalization (sum all cycles = 0)

$$D_t = Y_t - T_t \quad S'_{jan} = \frac{1}{N} \sum_{years} D_{jan} \quad S_{jan} = S'_{jan} - \frac{1}{m=12} \sum_{months} S'_{month_i}$$

- **Residual**: rest

$$R_t = Y_t - T_t - S_t$$



Decomposition methods

Negatives of basic method:

- Seasonal pattern identical in each cycle (year)
- Lose bounds of data in trends

STL Decomposition (Seasonal + Trend decomp using LOESS)

- LOESS: Locally Estimated Scatterplot Smoothing
- Localized regression (window) based on immediate neighbors
- Far points weighted less
- Weighted Least Squares
- Regression model: line + parabola
- For every data point, one regression model



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Forecasting



What is forecasting?

- **Predicting the future** using **historical** data and knowledge of future events that might affect our forecasts.
- **Example:**
 - **Goal:** to know which sleeping bag to bring to sleep comfortably at night. If nights will be cold, warm sleeping bag. Otherwise, a light sleeping bag.
 - **What do we have to forecast?:** the temperature at night.
 - **Set the horizon:** the camping trip will in one week from now and it will **last for one week**.
 - **Gathering data:**
 - Historical **daily** minimum temperature data.
 - Gather data on possible factors that can influence temperature, such as **humidity** and **wind speed**.



What is forecasting? (II)

- Example:
 - **How much data?**
 - Ideally, you would collect more than 1 year of data → yearly **seasonal** pattern or a trend.
 - **Frequency** → Daily forecasts? Hourly?
 - As we increase the frequency, we should need less **historical period** in order to have enough data. (As usual, not always)
 - **No clear answer.**



What is forecasting? (II)

- Example:
 - **Which Machine Learning model should we use?**
 - Seasonality → SARIMA.
 - If you have mix information (wind speed + humidity) → SARIMAX.
 - Large amount of data → Deep learning.
 - **Train and test sets:**
 - Test set will always be the **most recent** data points (or at least try).
 - Test set must be representative of the **forecasting horizon**.
 - Remove last seven days of data → our test set.
 - **Performance assessment**
 - Error metric → Mean Squared Error (MSE), Mean absolute percentage error (MAPE), etc.



What is forecasting? (III)

- Which is the main difference between forecasting and regression?
 - In both task we must predict some **continuous target**.
 - Time series have an **order** and it must be **maintained**.
 - It is possible to forecast time series **without** the use of **features** other than the time series itself.
 - E.g., task of predicting revenue based on ad spend:
 - Features: **amount spent on Google ads**, Facebook ads, and television ads → Regression model.
 - Features: **time point** and **revenue** in the given time point → Time series.



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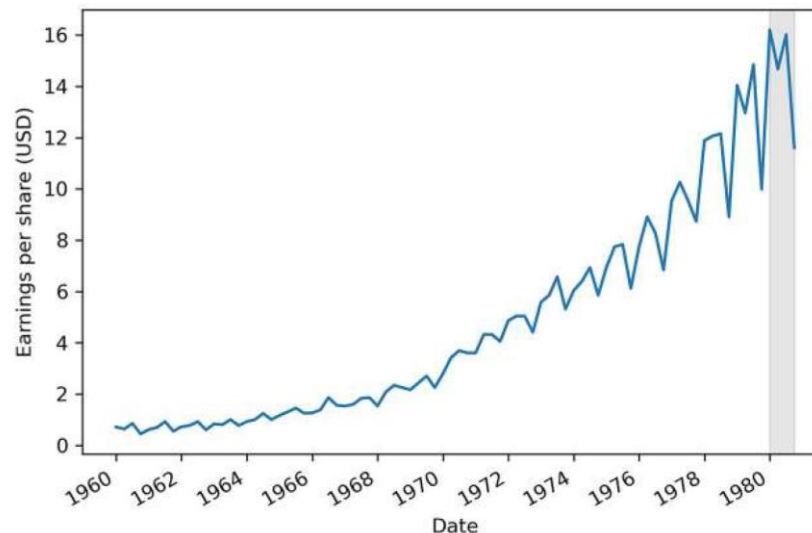


Forecasting – naive



A naive prediction of the future

- **Goal:** to set a **baseline** using a naive approach.
 - Trivial solution that uses **heuristics** or **simple statistics** to forecast.
 - Often require → **intuition** obtained by **visualizing** the data and detecting the **patterns**.
 - To compare it with more **complex approaches**.



Let's forecast the 4 quarters of 1980 (in grey) based on the previous temporal data.



A naive prediction of the future (II)

- Most simple statistic → **arithmetic mean**.
 - Mean of the values over a certain period → Assuming that future values will be equal to that mean.
 - E.g.: The average EPS between 1960 and 1979 was \$4.31. Therefore, I expect the EPS over the next four quarters of 1980 to be equal to \$4.31 per quarter.
- Heuristics → Forecast the **last recorded data point**.
 - E.g.: If the EPS is \$0.71 for this quarter, then the EPS will also be \$0.71 for next quarter.



A naive prediction of the future (III)

- **Cyclical pattern** → Repeat the pattern.
 - E.g.: If the EPS is \$14.04 for the first quarter of 1979, then the EPS for the first quarter of 1980 will also be \$14.04.
- How to **evaluate** which prediction is the best one?
 - **Out-of-sample forecasting** → making predictions for a period that was not taken into account when the model was developed.
 - E.g.: forecasting for the year of 1980 and testing our forecasts against the observed data in 1980.



Mean absolute percentage error (MAPE)

- Measure of **prediction accuracy** for forecasting methods that is **easy** to **interpret** and **independent** of the **scale** of our data.
 - Whether we are working with two-digit values or six-digit values, the MAPE will always be expressed as a **percentage**.
- The **percentage** of how much the forecast values **deviate** from the **observed** or actual values on average, whether the prediction was higher or lower than the observed values.
 - A_i is the actual value at point i in time.
 - F_i is the forecast value at point i in time.
 - n is the number of forecasts (e.g., test set).

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right| \times 100$$



A naive prediction of the future (IV)

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Random Walks - definition



Random Walks

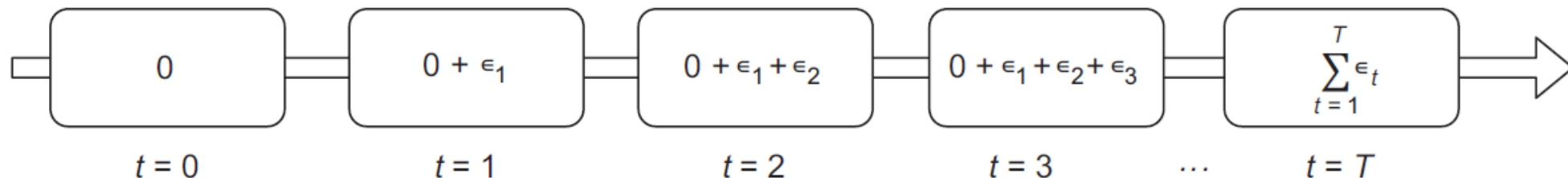
- A process in which there is an **equal chance** of going **up** or **down** by a **random number**.
 - Usually observed in **financial** and **economic** data.
- **Long periods** where a **positive** or **negative trend** can be observed → often accompanied by **sudden changes** in **direction**.
- Mathematically expressed as:
 - y_t is the value at the present time t .
 - C is a constant (drift).
 - y_{t-1} is the value at the previous time step $t-1$.
 - ϵ_t is a random number, also known as **white noise**.

$$y_t = C + y_{t-1} + \epsilon_t$$



Simulating a Random Walk

- We are going to **simulate** a **random walk** in order to understand this phenomenon.
- To do so, we are going to follow these assumptions:
 - Constant C is 0.
 - $y_0 = 0$
 - The value at $t=1$, y_1 will be equal to the previous value plus **white noise**.



The construction of the simulated random walk.



Simulating a Random Walk (II)

- [Colab.](#)
- We know is a random walk because we have simulated it. However...

How can we identify whether our time series is a random walk or not?



Identifying a random walk

- A random walk is a series whose *first difference* is *stationary* and *uncorrelated* → meaning that the process moves completely at random.

What is a stationary series?
What is an uncorrelated series?
What is the first difference?



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Random Walks - stationarity



Stationarity

- Those time series whose **statistical properties** do **not change over time**.
 - **Constant mean.**
 - **Constant variance**
 - **Constant autocovariance:** value in t vs value in $t-k$ should only depend on k value (lag), not on t (when they occur) (e.g.: temps vs stock market)
- There are **some models** which **only work** with **stationary** time series.
 - E.g.: moving average models, autoregressive models or autoregressive average models.
 - Non-stationary datas' statistical properties changes over time, affecting model properties.
 - Model parameters must adapt to these changes, complicating the derivation of future values from past values.
 - Constantly changing coefficients make forecasting unreliable.



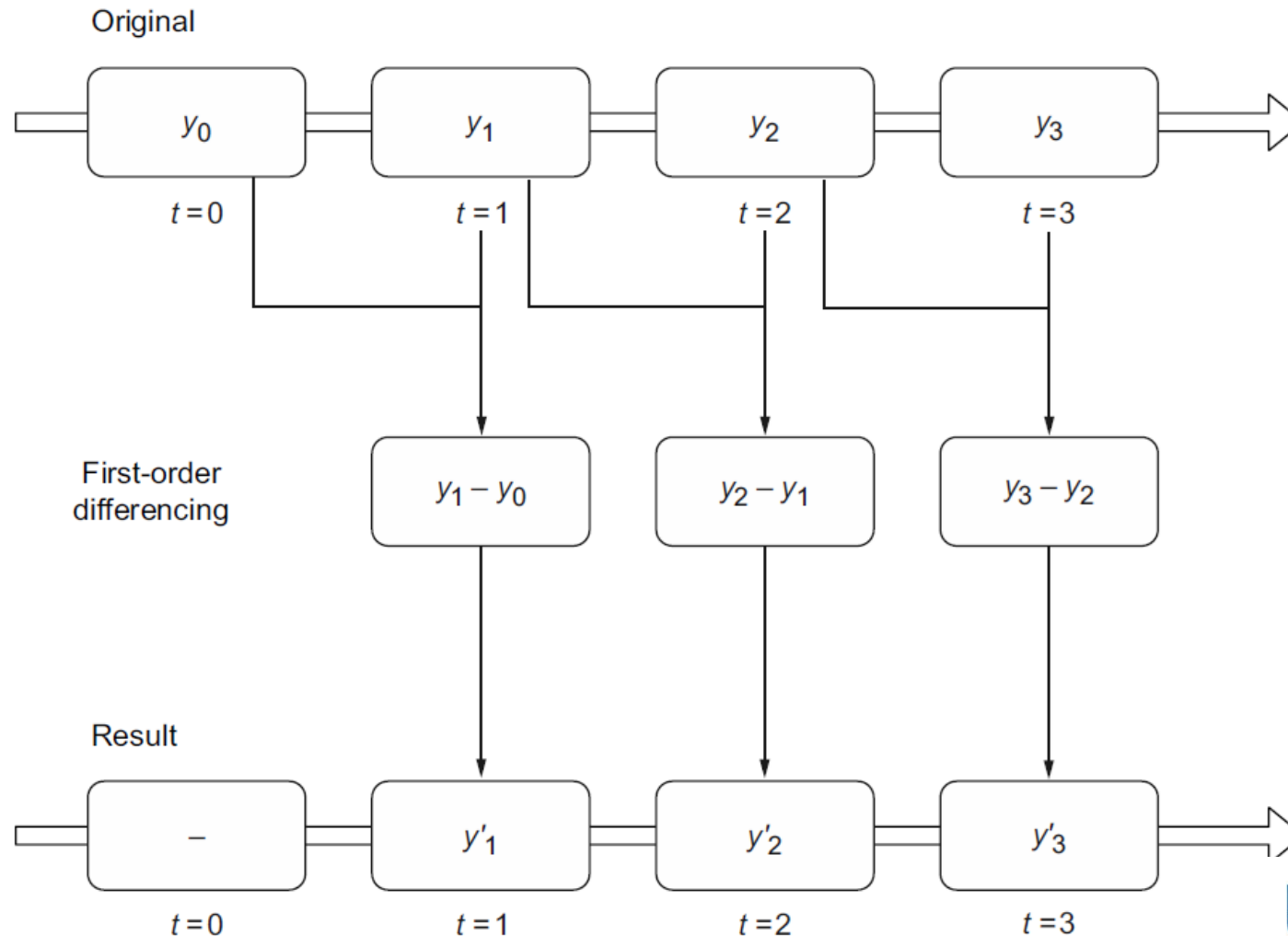


Stationarity (II)

- What if... our time series is not stationary?
 - We can **transform** it and make it **stationary**.
 - A transformation is simply a **mathematical manipulation** of the data that **stabilizes** its **mean** and **variance**.
- The simplest transformation → **Differencing**.
 - It stabilizes the mean, **removing** or **reducing** the **trend** and **seasonality** effects.
 - Involves calculating the series of changes from one timestep to another.
 - **Logarithms** are used to help stabilize the **variance** (for exp cases).
- After it has been transformed, we have to **undo** the **transformation** through an ***inverse transform***.
 - E.g., if we apply a log transformation to the data, we have to raise the forecast value to the power of 10 → So that predictions make sense in the original context.



Stationarity - Differencing



The differencing transformation. This is a first order differencing, however, the process can be repeated more times.



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Testing for stationarity

- Once the **first transformation** is applied, we have to check if the time series is stationary or not.
 - If not, another transformation should be performed (**second order transformation**).
- **Dickey-Fuller (ADF) test.**
 - Verifies the following **null hypothesis**: there is a **unit root present** in the time series (the time series is **not stationary**).
 - The **alternative hypothesis**: there is **no unit root** (the time series is **stationary**).
 - The outcome of this test is a **negative number** → the more negative, stronger the rejection of the null hypothesis.
 - If the **p-value** is **less** than **0.05** we **reject** the **null hypothesis** → time series is **stationary**.



Testing for stationarity - ADF

Intuition:

- Let's consider a very simple time series where the present value y_t only depends on its past value y_{t-1} subject to a coefficient α_1 , a constant C , and white noise ϵ_t .
 - C is a constant.
 - α_1 is the root of the time series.
 - ϵ_t error that we cannot predict.
- The time series will be stationary if $1 > \alpha_1 > -1$.

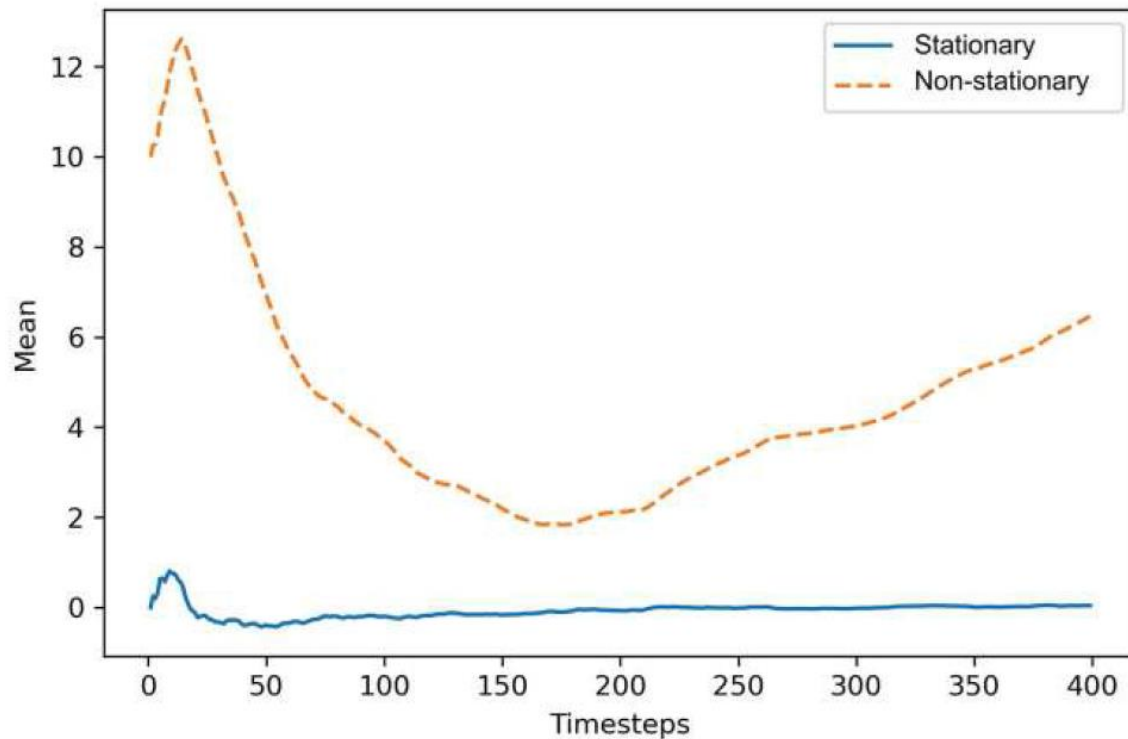
$$y_t = C + \alpha_1 y_{t-1} + \epsilon_t$$



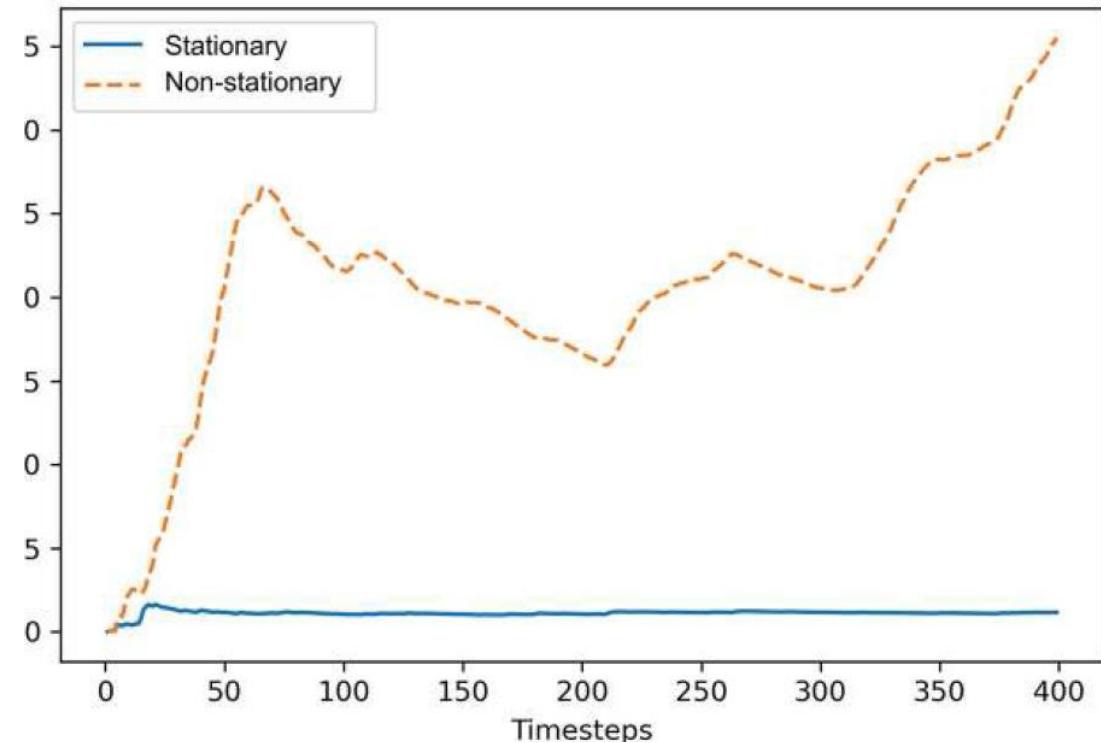
Testing for stationarity - ADF (ii)

$$y_t = 0.5y_{t-1} + \epsilon_t$$

$$y_t = y_{t-1} + \epsilon_t$$



Simulated time series. Mean of stationary and non-stationary series. The mean of the non-stationary series depends on the time.



Simulated time series. Mean of stationary and non-stationary series. The variance of the non-stationary series depends on the time.



Identifying a random walk (II)

- A random walk is a series whose *first difference* is *stationary* and *uncorrelated* → meaning that the process moves completely at random.

STATIONARY + UNCORRELATED



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Random Walks - autocorrelation



Autocorrelation function (ACF)

- Used to **analyze** if we are dealing with a **random walk or not**.
- **Correlation** → measures the extent of a linear relationship between two variables.
- **Autocorrelation** → measures the linear relationship between **lagged values** of a time series.
 - ACF reveals how the **correlation** between any two values **changes** as the **lag increases**.
 - The **lag** is the **number of timesteps separating two values**.



Understanding Autocorrelation in Time Series

- **Autocorrelation Coefficient:** Measures the linear relationship between current time series values and their previous values at a given lag.
 - **Example:** Coefficient between y_t and y_{t-1} is denoted as r_1 for lag 1, and between y_t and y_{t-2} as r_2 for lag 2.
- **Plotting the ACF function:**
 - The autocorrelation coefficient (dependent variable, y axes) vs. the lag (independent variable, x axes).
 - At lag 0, the coefficient is always 1, indicating a perfect linear relationship with itself.

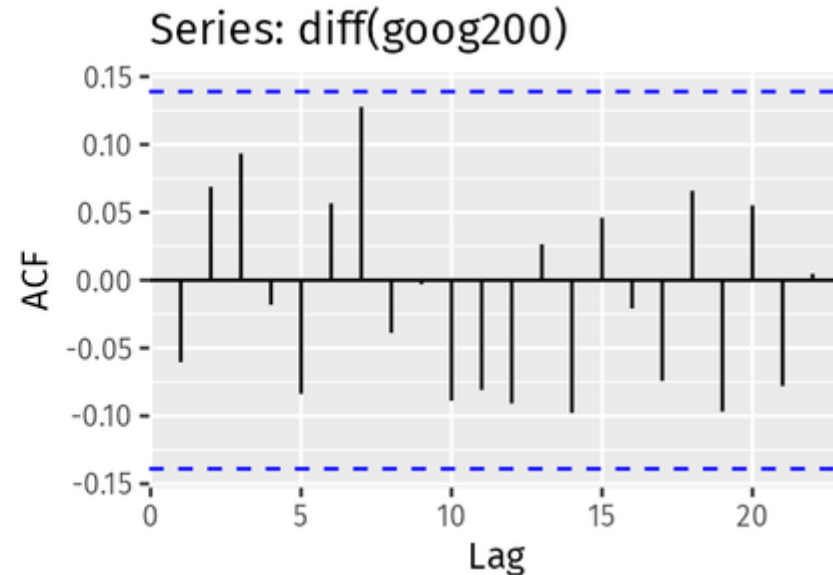
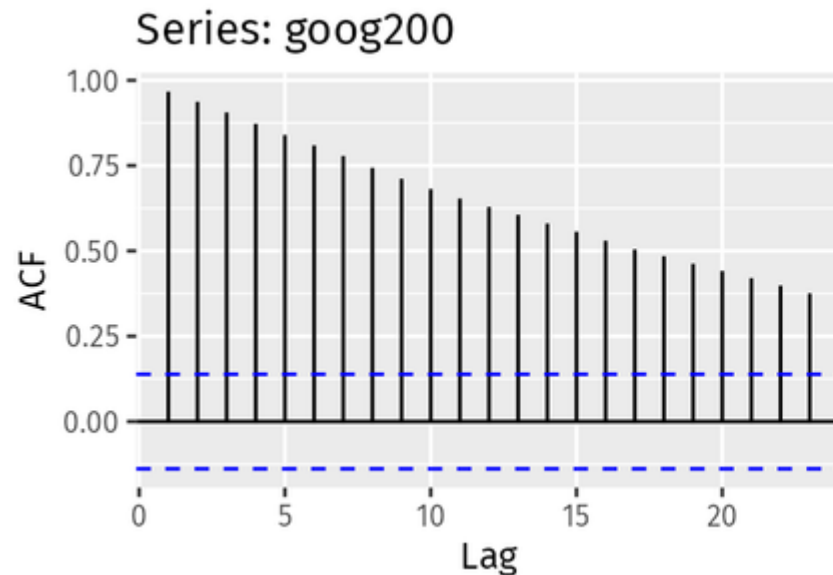
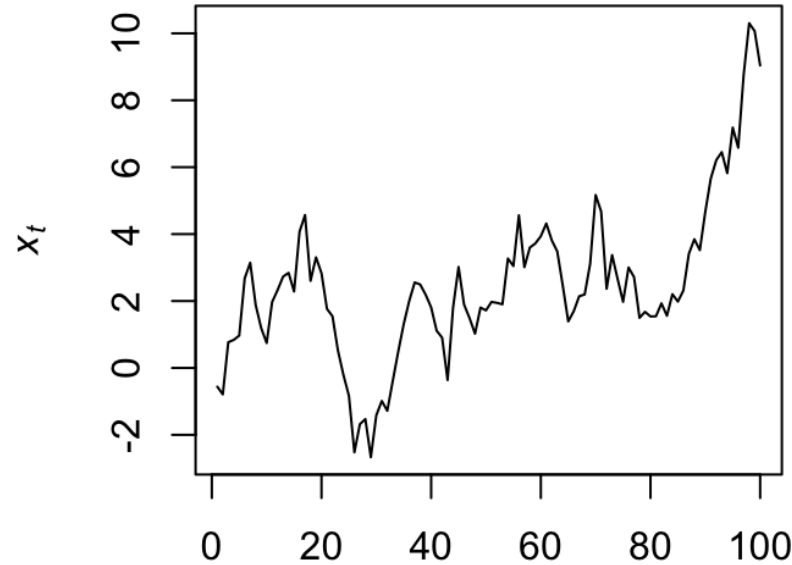


Autocorrelation and Data Characteristics

- **ACF and Trends:**
 - High coefficients for short lags that decrease linearly with longer lags, indicating the presence of a trend or a random walk (both non-stationary).
- **ACF and Seasonality:**
 - Cyclical patterns in the ACF plot signal seasonal data.
- **Importance of ACF:**
 - For non-stationary data, the ACF plot might not reveal more than the time evolution of the process.
 - In stationary data, ACF helps identify random walks, enhancing our understanding of the data's underlying patterns.

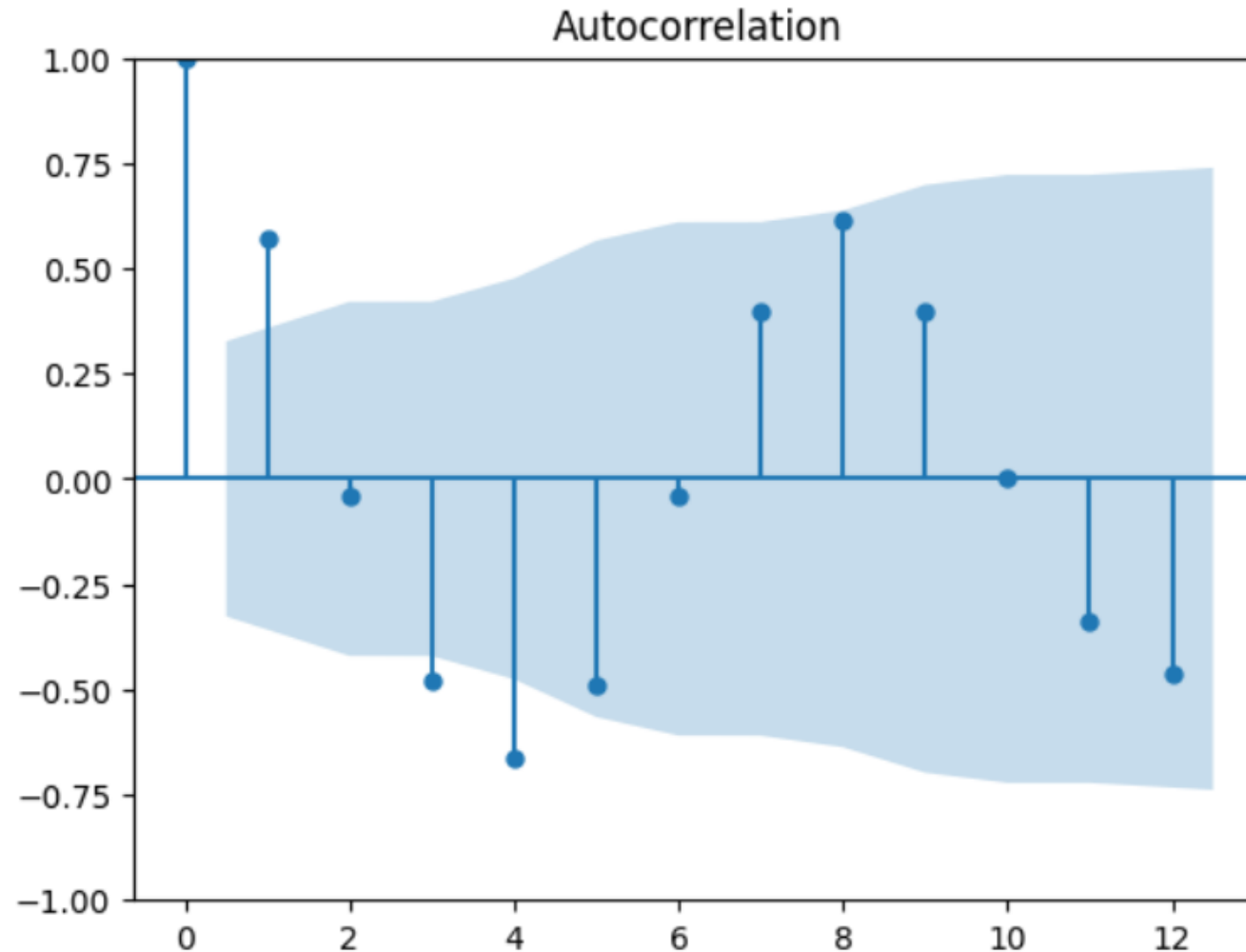


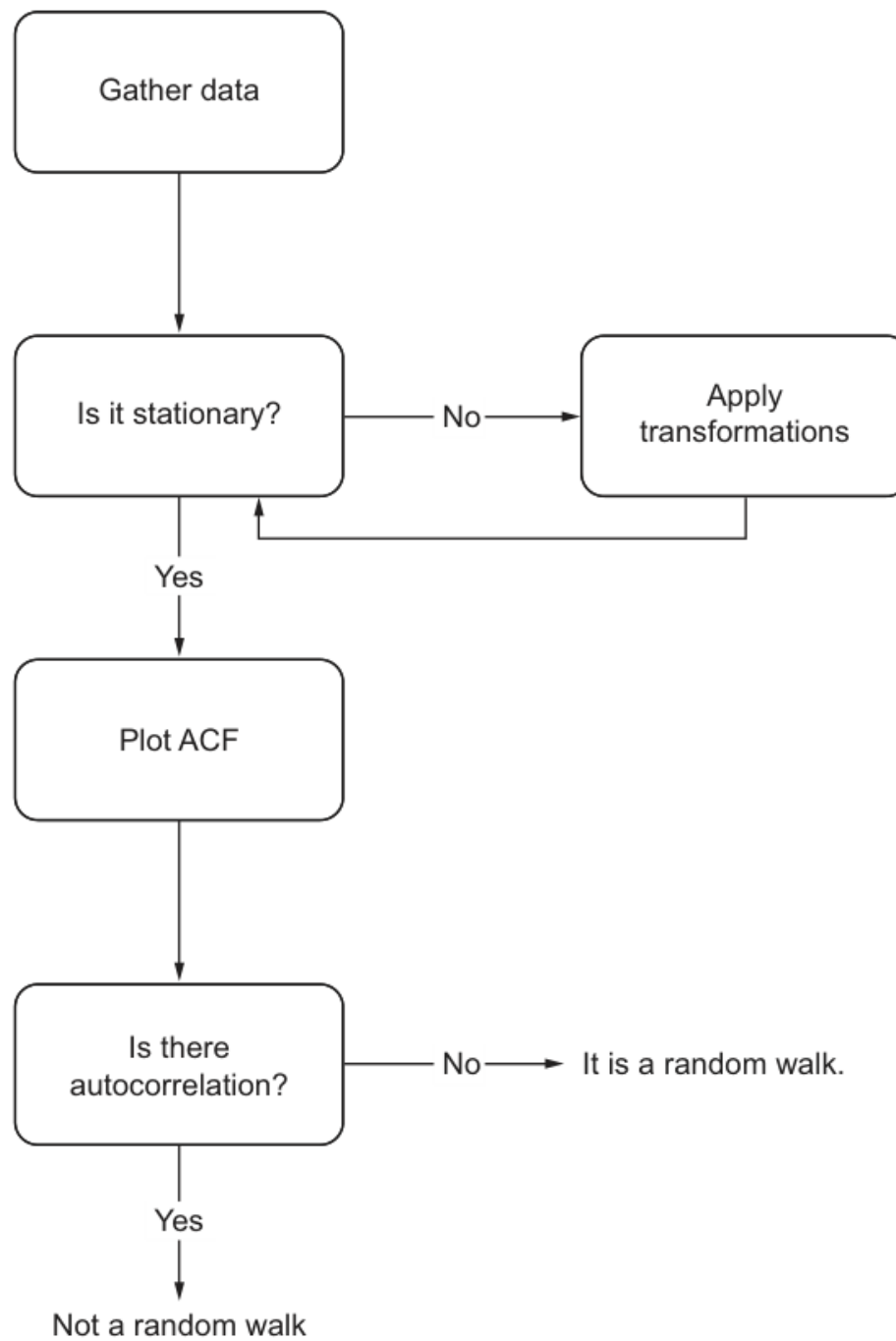
Autocorrelation and Data Characteristics





Autocorrelation and Data Characteristics







Identifying a random walk (III)

- [Colab.](#)



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Random Walks - forecasting



Forecasting Random Walks

- **What we know so far:** a random walk takes **random steps** as time progresses.
 - How can we forecast random walks?
 - Predicting a random change is impossible → unless we predict a random value ourselves (not ideal of course).
- We use **naive** forecasting **methods**.
 - Since the values change randomly → **no statistical learning** can be used.
- Ideally, when dealing with random walks, we will only forecast the next timestep.



Random Walk forecasting on a long horizon

- Mean of the whole training set.
- Last known value of the train set.
- Drift method:
 - A **modification** of predicting the **last known value**.
 - The **values increase** or **decrease** over **time**.
 - The rate at which values will change in the future is equal to the one seen in the training set.
 - Calculate the **slope** between the **first and last value** and extrapolate this **straight line** into the future.



Drift Method

- The slope can be calculated **dividing** the **change** in the **y-axis** by the **change** in the **x-axis**.
 - Following the example of the synthetic dataset:
 - The change in the **y-axis** is the **difference** between the **last value** of our random walk y_f and its **initial value** y_i .
 - The change in the **x-axis** is equivalent to the number of **timesteps minus 1**.

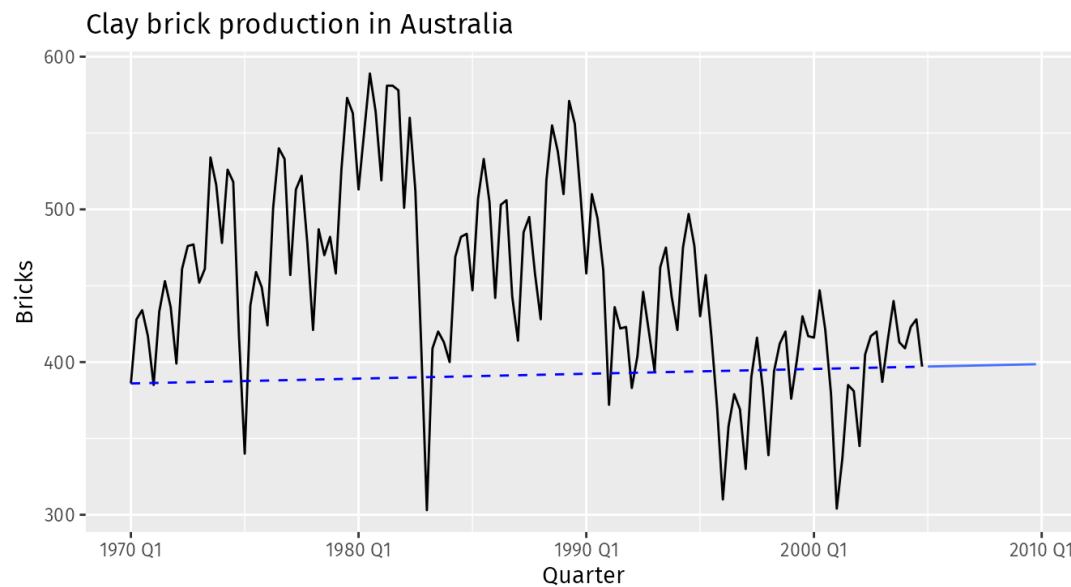
$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_f - y_i}{\# \text{ timesteps} - 1}$$

- It states that the value of the forecast is linearly dependent on the timestep, value of the drift and the initial value of the timestamp.

$$\text{forecast} = \text{drift} \times \text{timestep} + y_i$$



Drift Method



Clay brick production in Australia. [Source.](#)



NASDAQ. [Source.](#)

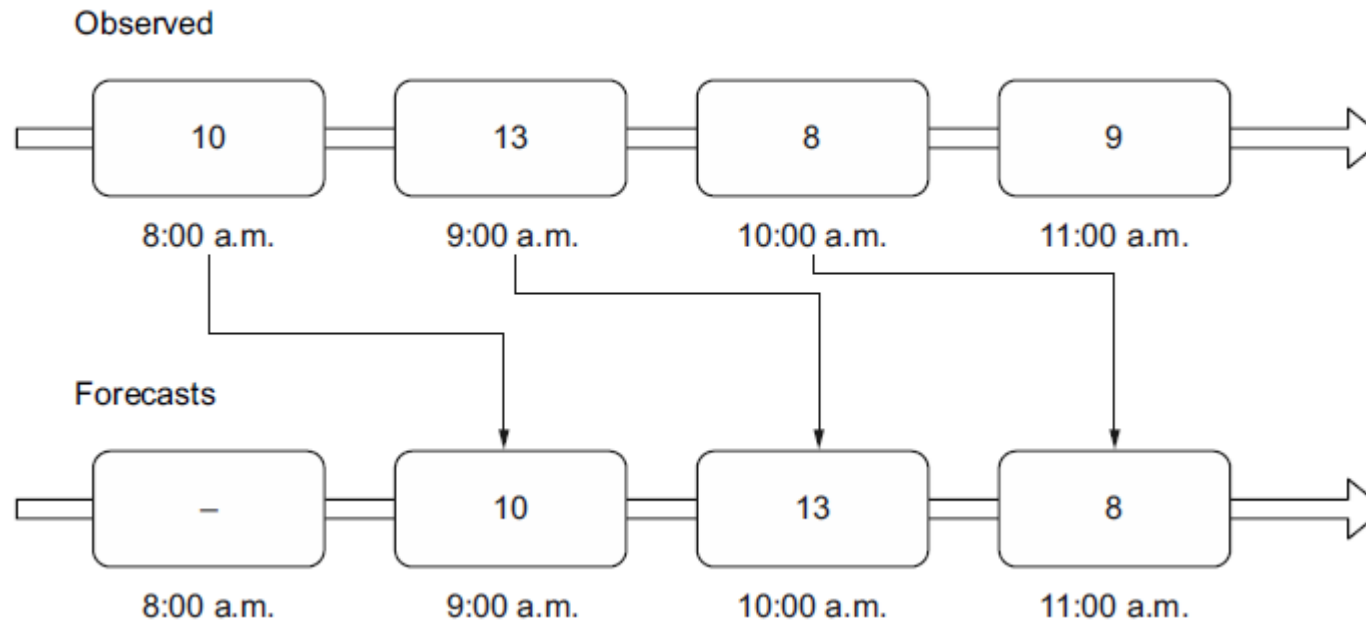


Random Walk forecasting on a short horizon (next timestep)

- **Objective:** Forecast the next time step in a random walk scenario.
- **How:** Predicting the last known value for the next timestep.
- **Intuition:** The future value of a random walk is essentially the **past value plus white noise**, making this approach reasonable.
- **Implementation:**
 - Start with the **initial observed value** to predict the next timestep.
 - Upon **observing** a new value, **use this latest value to forecast the subsequent timestep**.
 - **Repeat** the process for future timesteps, always using the last **recorded value** as the forecast for the next step.
- **Outcome:** The **forecast** will typically be off by a **random number**, reflecting the inherent **unpredictability** of a random walk with white noise.



Random Walk forecasting on a short horizon (next time step) (II)



The observed value at 8:00 a.m. is used to forecast the value for 9:00 a.m., the actual value observed at 9:00 a.m. is used to forecast the value at 10:00 a.m., and so on.

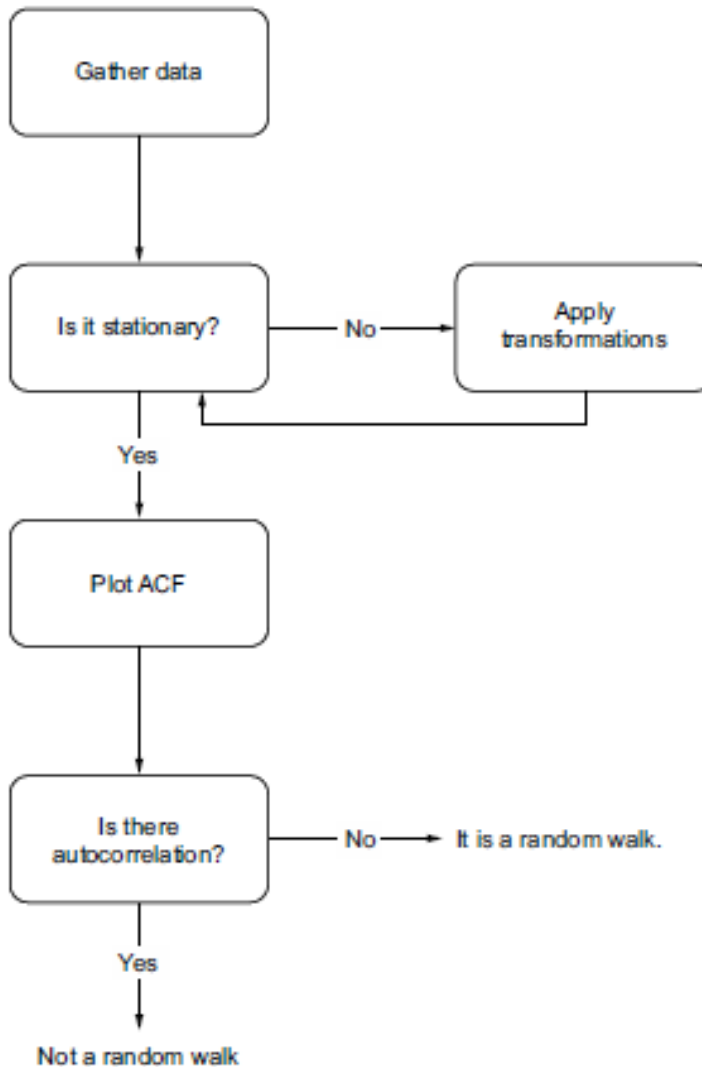


Random Walk forecasting on a short horizon (next time step) (III)

- **E.g.:** Consider a simplified example where a stock price follows a random walk. If the stock price today is \$100, the best forecast for tomorrow's price is also \$100, not because we expect it not to change, but because we have no basis to predict whether it will go up or down. Any change from \$100 would be due to random factors that are, by definition, unpredictable.
- [Colab.](#)



Non random walk time series





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Moving Average process (MA)



Non random walk time series (II)

- Stationary + Uncorrelated = Random Walk
- Stationary + Autocorrelation = **????**
 - A time series that can be approximated by:
 - **A moving average model MA.**
 - An autoregressive model AR.
 - Autoregressive moving average model ARMA.



Modeling a moving average process

- **Definition:** a process where the current value is linearly influenced by current and past error terms.
- **Characteristics of error terms:** error terms are mutually independent and normally distributed, resembling *white noise*.
- **Moving Average Model Notation:** the model is denoted as **$MA(q)$** , where **q** represents the **order** of the model.
- **Formula and components:**
$$y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \dots + \theta_q\epsilon_{t-q}$$
 - Linear combination of the mean of the series μ , the **present** error term ϵ_t , and **past** error terms ϵ_{t-q} .
 - **Magnitude** of the **impact** of **past errors** on the present value is controlled using the θ_q .

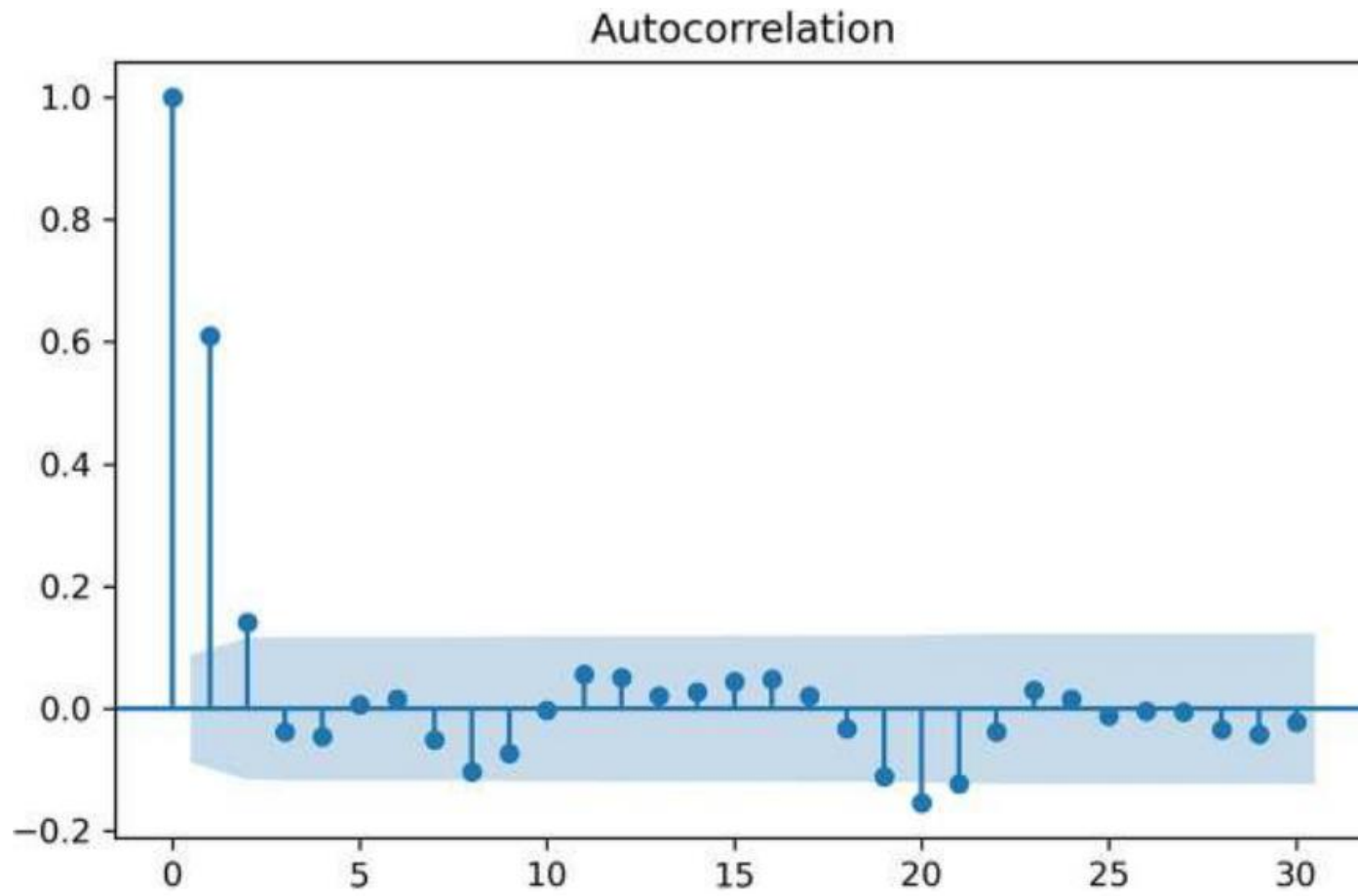


Modeling a moving average process (II)

- The order q of the $MA(q)$ process affects the **number of past error terms** that must be included in the model.
- The larger q is, the **more past errors** terms **affect** the present value.
- **IMPORTANT:**
 - To **determine** the **order** of the moving average process to fit the appropriate model → we need to **plot** the **ACF**.

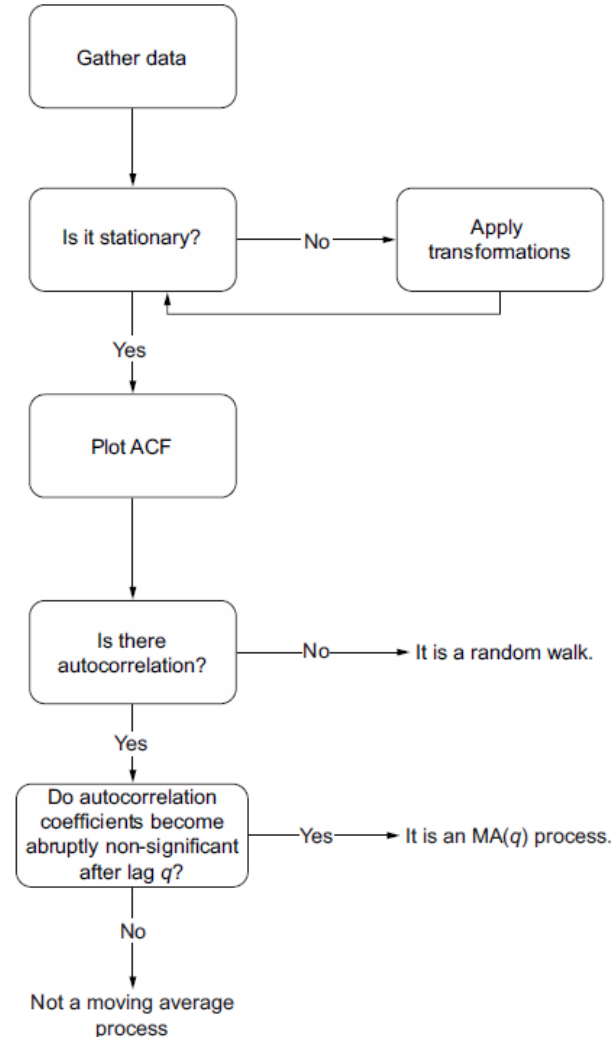


Modeling a moving average process (II)





Identifying the order of a moving average process





Identifying the order of a moving average process (II)

- [Colab.](#)



Forecasting a moving average process

- Once the **order q** of the moving average process is **identified**:
 - The **forecasts** must be done on a **stationary** time series.
 - **Forecast up until q values'** time window.
Unable to predict more values correctly.
 - Once the forecasts have been computed → have to **inverse-transform** the forecasts (undo the differencing).
 - We need to recover the initial value y_0 which was removed in the differencing process.

$$y_1 = y_0 + y'_1 = y_0 + y_1 - y_0 = y_1$$

- Can be obtained using a cumulative sum of the differenced values.

$$y_2 = y_0 + y'_1 + y'_2 = y_0 + y_1 - y_0 + y_2 - y_1 = (y_0 - y_0) + (y_1 - y_1) + y_2 = y_2$$



Forecasting a moving average process (II)

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AutoRegressive process
(AR)



Modeling on autoregressive process

- An autoregressive process establishes that the **output variable depends linearly** on its **previous values** → a regression of the variable itself.
- Denoted as **AR(p)** being p the order.
 - Present value y_t is a linear combination of a constant C , the present error term ϵ_t (white noise) and the past values of series y_{t-p} .
 - Magnitude of influence of the past values on the present value is denoted as ϕ_p .

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

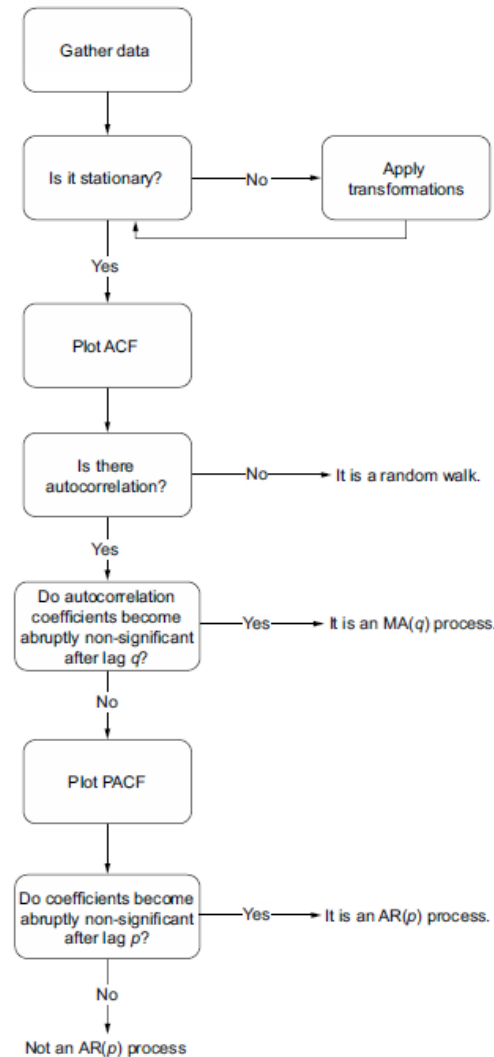
- The order p determines the number of past values that affect the present value.

$$y_t = C + \phi_1 y_{t-1} + \epsilon_t$$

Autoregressive process of order 1.



Identifying the order a stationary autoregressive process





Partial AutoCorrelation Function (PACF)

- When the **ACF** plot exhibits a pattern of an **exponential decay** → we may have an **autoregressive process** → **partial** autocorrelation function (**PCAF**) must be plotted and analyzed.
 - Autocorrelation measures the linear relationship between lagged values of a time series.
 - What is partial autocorrelation then?

$$y_t = 0.33y_{t-1} + 0.50y_{t-2}$$

- With ACF we are measuring the correlation between y_t and y_{t-2} .
- We can see that y_{t-2} has influence in y_{t-1} .
 - Why? Because we are in an AR(2) process and each value depends on the previous two values.
- However, we are **NOT** measuring the **true impact** of y_{t-2} on y_t .

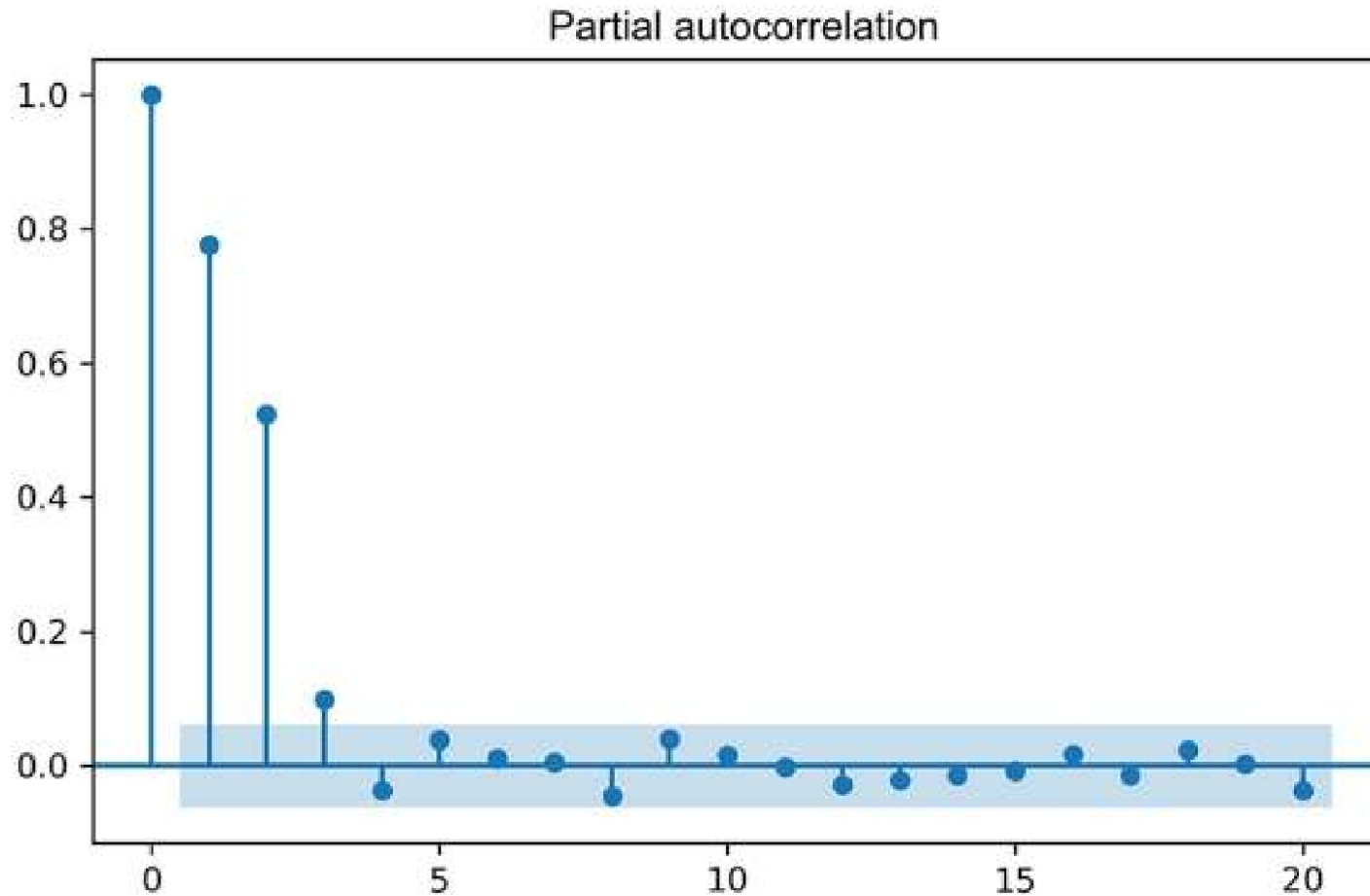


Partial AutoCorrelation Function (PACF) (II)

- We are **NOT** measuring the **true impact** of y_{t-2} on $y_t \rightarrow$ To do it, we need PCAF.
- What is PCAF or Partial AutoCorrelation?
 - Measures the correlation between lagged values in a time series when we **remove the influence of correlated lagged values in between.**
 - We can **plot the partial autocorrelation function** to determine the **order** of a **stationary AR(p)** process.



Partial AutoCorrelation Function (PACF) (II)





Partial AutoCorrelation Function (PACF) (II)

- [Colab.](#)



Forecasting an autoregressive process

- Once the **order p** of the autoregressive process is **identified**:
 - The **forecasts** must be done on a **stationary** time series.
 - We can try several **naive forecasting** techniques (as **baselines**) and the second-order moving average model.
 - Once the forecasts have been computed → have to **inverse-transform** the forecasts (undo the differencing).
 - We need to recover the initial value y_0 which was removed in the differencing process.

$$y_1 = y_0 + y'_1 = y_0 + y_1 - y_0 = y_1$$

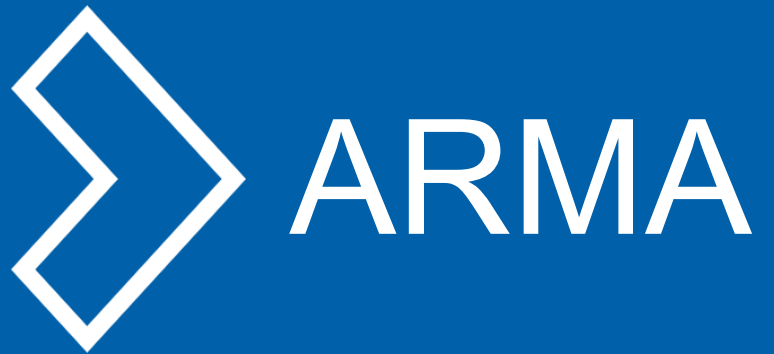
- Can be obtained using a cumulative sum of the differenced values.

$$y_2 = y_0 + y'_1 + y'_2 = y_0 + y_1 - y_0 + y_2 - y_1 = (y_0 - y_0) + (y_1 - y_1) + y_2 = y_2$$



Forecasting an autoregressive process (II)

- [Colab.](#)





Modeling complex time series

- **Moving Average** process $\rightarrow$ $MA(q)$.
 - The **present value** is linearly dependent on the **mean**, the **current error term** and **past error terms**.
 - Order **q** can be inferred using the **ACF** plot.
 - If the ACF plot shows a **slowly decaying pattern** or **sinusoidal pattern** $\rightarrow$ **may** indicate **presence** of an **autoregressive** process.
- **AutoRegressive** process $\rightarrow$ $AR(p)$.
 - The **present value** is linearly dependent on its **own past value**.
 - Order **p** can be inferred using the **PACF** plot.
 - Partial autocorrelation **coefficients** will be **significant** up until **lag p** only.



Modeling complex time series (II)

... we still have slowly decaying pattern or a sinusoidal pattern after ACF or PACF?

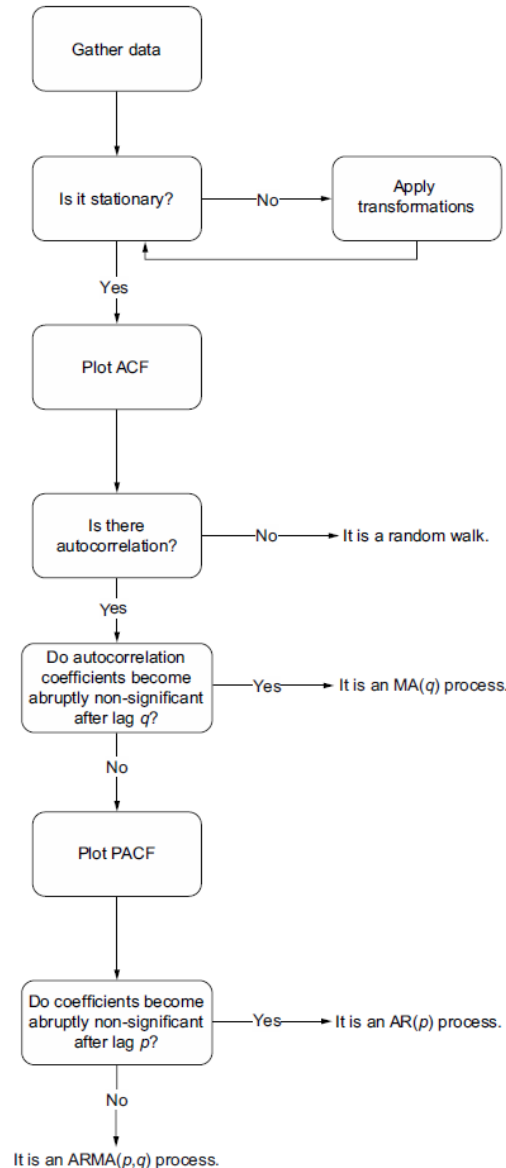


Modeling complex time series (III)

AutoRegressive Moving Average (ARMA).



Modeling complex time series (IV)





Modeling on AutoRegressive Moving Average (ARMA)

- Combination of MA and AR.
 - The present value is linearly dependent on its own previous values and a constant (AR).
 - **AND**, on the mean of the series, the current error term, and past error terms (MA).
- Denoted as **ARMA(p,q)** where **p** is the order of the autoregressive portion, and **q** is the order of the moving average portion.
 - Order **p** determines the number of past values that affect the present value.
 - Order **q** determines the number of past error terms that affect the present value.

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

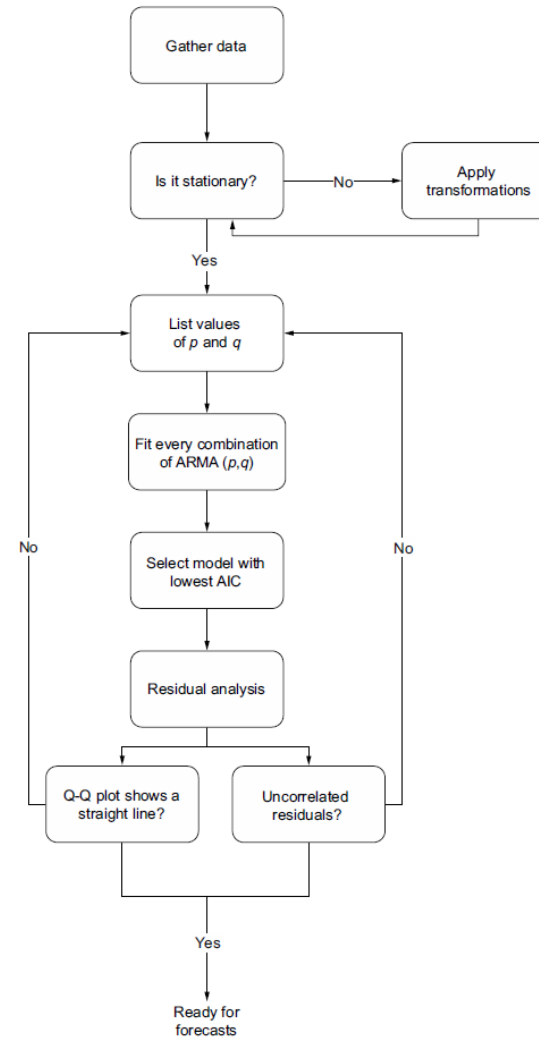


Modeling on AutoRegressive Moving Average (ARMA) (II)

- The **methods** for **identifying** q and p values **change** when dealing with an **ARMA** process.
- Let's simulate an ARMA process and see what happens if we use ACF and PACF.
 - [Colab](#).



Modeling on AutoRegressive Moving Average (ARMA) (III)





Select the model with AIC

- The optimal model will be chosen using the **Akaike Information Criterion (AIC)**.
- AIC estimates the quality of a model relative to the others → Quantifying how much information is lost when a model is fitted to the data.
 - Less information lost, the lower the AIC value and the better the model.
- The AIC is a function of the number of parameters k in a model and the maximum value of the likelihood function:

$$\text{AIC} = 2k - 2 \ln(\hat{L})$$

$$\ell(\theta, \sigma^2) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T \epsilon_t^2$$



Select the model with AIC (II)

- The number of estimated parameters k is directly related to the order (p,q) of an ARMA(p,q) model.
 - E.g.: if we fit an ARMA(2,2) model, then we have 2 + 2 parameters to estimate.
 - Arma (3,4), 3 + 4 parameters to estimate.
 - Therefore, as the order increases, the complexity of the model increases (since the number of parameters also increases).
 - AIC keeps a balance between underfitting and overfitting.
 - The larger k is, more will be penalized the metric.
- Likelihood function measures the goodness of fit of a model.
 - **Distribution function:**
 - Measures the probability of observing a data point, given fixed model parameters.
 - **Likelihood function:**
 - Estimates the likelihood that different model parameters will generate the observed data, given a set of observed data.



Select the model with AIC (III)

- Let's analyze how AIC works in a [Colab](#).



Residual analysis

- **After** using **AIC** to find the best (p, q) parameters and therefore the best model **relative** to all others we have fit (all those permutations) → we **MUST** measure its **absolute** quality analyzing model's **residuals**.
- To do so, we must answer two questions:
 - **Does the Q-Q plot show a straight line?**
 - **Are the residuals uncorrelated?**
- If the **answer to both questions** is **YES**, we can start forecasting.
 - If not, **restart the process** and try new (p, q) combinations.



What are residuals?

- The **difference** between the **predicted** values and the **actual** values.

$$y_t = 0.33y_{t-1} + 0.9\epsilon_{t-1} + \epsilon_t$$

Our ARMA(1,1) time series.

$$\hat{y}_t = 0.33y_{t-1} + 0.9\epsilon_{t-1}$$

Our ARMA(1,1) model.

$$\text{residuals} = 0.33y_{t-1} + 0.9\epsilon_{t-1} + \epsilon_t - (0.33y_{t-1} + 0.9\epsilon_{t-1})$$

Subtraction between the "truth" and the forecast.

$$\text{residuals} = \epsilon_t$$



What are residuals?

- In a perfect situation the **residuals** of a model are **white noise**.
- Indicates that the **model** has **captured** all **predictive information** → there is **only** a **random fluctuation** left that cannot be modeled.
 - The residuals must be **uncorrelated** and have a **normal distribution** to conclude that we have a good model for making forecast.
- Two analysis:
 - **Qualitative**: studying the **Q-Q plot**.
 - **Quantitative**: determines whether our **residuals** are **uncorrelated**.



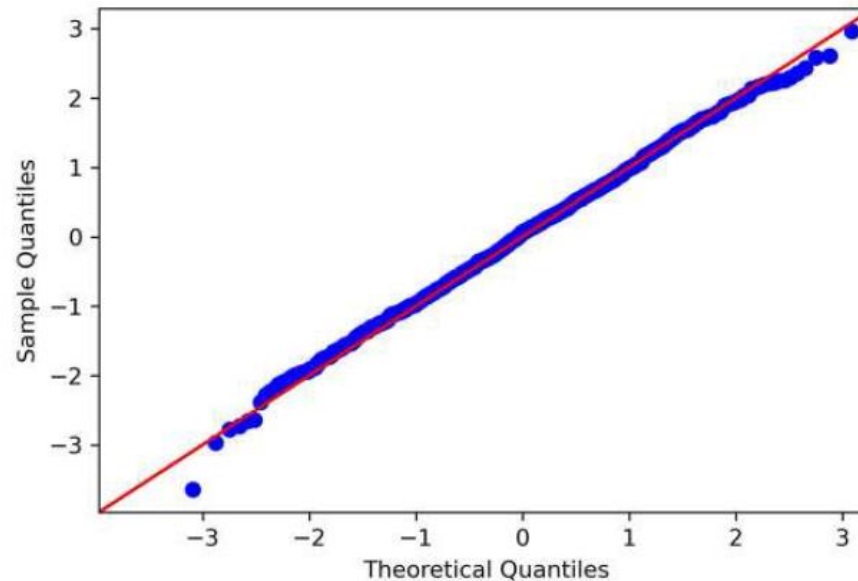
Qualitative Analysis: Studying the Q-Q plot

- Qualitative analysis.
- **Graphical** tool to verify our hypothesis: model's **residuals** are **normally distributed**.
- The Q-Q plot is built by plotting the quantiles of our **residuals** on the **y-axis** against the of a **theoretical normal** distribution on the **x-axis**.
 - Why? Because we want the **residual** to be **similar** to **white noise**, which is normally distributed.



Qualitative Analysis: Studying the Q-Q plot (II)

- If **both distributions** are similar (the distribution of the residuals is close to a normal distribution) → means that the model is good fit for the data.

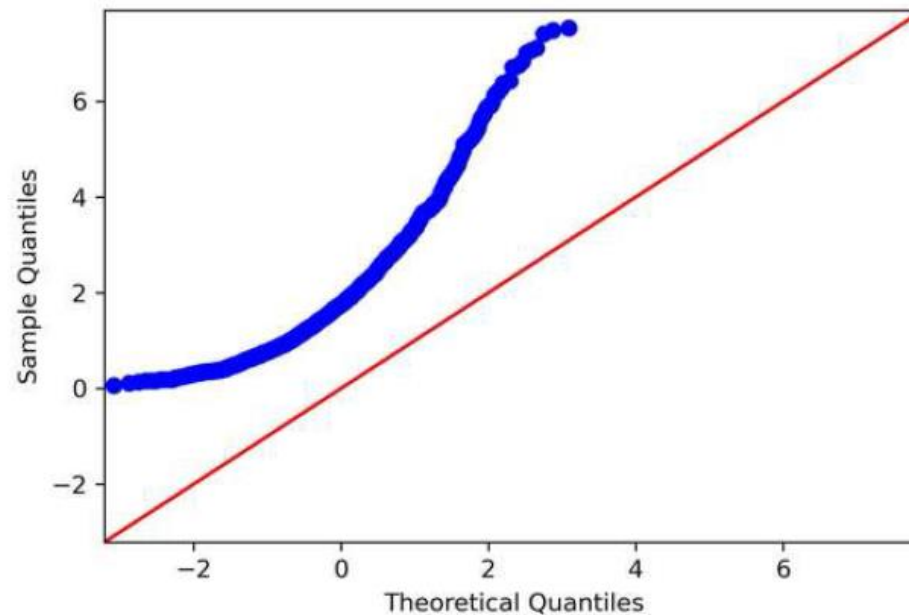


Y-axis quantiles coming from the residuals. X-axis coming from a normal distribution. If we have a straight line on $y=x$ means that residuals are close to a normal distribution.



Qualitative Analysis: Studying the Q-Q plot (III)

- On the other hand, if the **residuals** are **not close** to a **normal distribution**, it will generate a curve that departs from $y=x$.



A Q-Q plot of residuals that are not close to a normal distribution.



Qualitative Analysis: Studying the Q-Q plot (IV)

- Even though is a fast method for assessing the quality of the model → it is a subjective analysis.
 - Quantitative approach is needed → Ljung-Box test.



Quantitative Analysis: Applying the Ljung-Box test

- The Ljung-Box test is a statistical test that tests if the **autocorrelation** of a **group of data** is significantly **different** from **0**.
 - The **null hypothesis** states that the **data is independently distributed** (therefore **no autocorrelation**).
- In this case, we will **apply** it to the **model residuals** → to test whether they are **similar to white noise** or no.
 - If p-value is **larger** than **0.05** → **cannot reject the null hypothesis** → residual are independently distributed → no autocorrelation and therefore similar to white noise.
 - If p-value smaller than **0.05** → **reject the null** → data is not independently distributed, correlation and therefore we **can not affirm residuals** are **similar to white noise**.



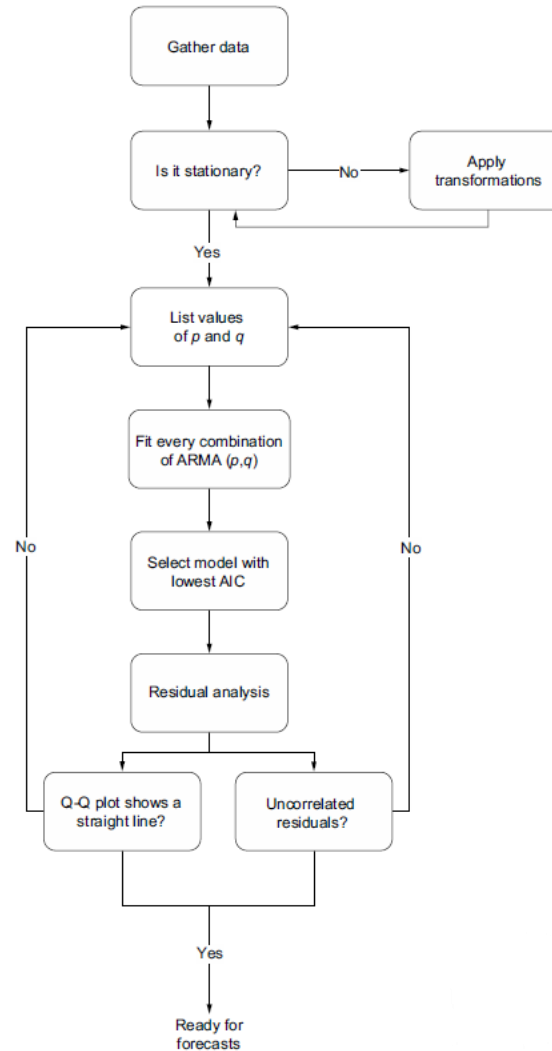
Quantitative Analysis: Applying the Ljung-Box test

- [Colab.](#)



General Modeling Procedure

- [Colab.](#)







Forecasting non-stationary time series

- Adding a new component to $ARMA(p,q)$ we will be able to forecast non-stationary time series → ***Integration order (d)***.
- **AutoRegressive Integrated Moving Average** → $ARIMA(p,d,q)$.
 - We can use non-stationary time series.
 - We do not have to model on difference data.
 - We do not have to inverse transform the forecasts.

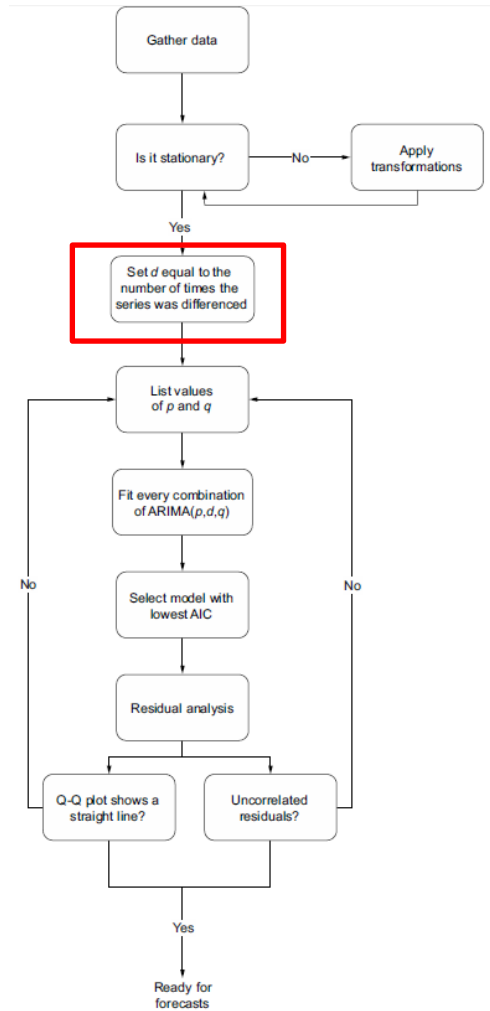


AutoRegressive Integrated Moving Average (ARIMA)

- Combination of an autoregressive process $\mathbf{AR}(p)$, integration $\mathbf{I}(d)$, and the moving average process $\mathbf{MA}(q)$.
- The present value is dependent on past values, coming from $\mathbf{AR}(p)$ portion, and past errors, coming from the $\mathbf{MA}(q)$ portion.
 - However, in this case instead of using the original series, denoted as y_t , the ARIMA process uses the differenced series (which may have been differenced more than once) y'_t .
- p determines how many lagged values of the series are included in the model. q determines how many lagged error terms are included.
- The order d is defined as the **order of integration** → which is just the **reverse of differencing** → the number of time a series has been differenced to become stationary.



AutoRegressive Integrated Moving Average (ARIMA)





AutoRegressive Integrated Moving Average (ARIMA)

- Therefore, in the general modeling procedure we need to add an additional step to calculate **how many times a time series must be differenced** to become stationary.
 - This is the **d** we are looking for.
 - A time series **rarely** needs to be differenced **more than twice**.
- What happens when $d=0$?
 - The model will be **equivalent** to **ARMA(p,q)**.
 - $ARIMA(1,0,1) == ARMA(1,1)$



AutoRegressive Integrated Moving Average (ARIMA)

- [Colab.](#)



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SARIMA



Seasonal AutoRegressive Integrated Moving Average (SARIMA)

- Includes **seasonal patterns** in time series.
- Similar to creating 2 ARIMA, one in a classical way and another for seasonal data. (mixed by multiplicative operation)
- Adds **another** set of **parameters** allowing us to consider periodic patterns:
 - $(p,d,q)(P,D,Q)_m \rightarrow$ seasonal parameters.
- ***m*** stands for **frequency**: number of observations per **cycle**.
If the length of cycle is 1 year:
 - Data recorded annually $\rightarrow m=1$.
 - Data recorded quarterly $\rightarrow m=4$ (four quarters in a year/cycle).
 - Data recorded monthly $\rightarrow m=12$.
 - Data recorded weekly $\rightarrow m=52$.



Seasonal AutoRegressive Integrated Moving Average (SARIMA)

- And if the data is collected in daily or sub-daily basis?
 - Weekly seasonality: $m=7$ (seven observations in a full cycle).
 - Yearly seasonality: $m=365$.

Data collection	Frequency m				
	Minute	Hour	Day	Week	Year
Daily				7	365
Hourly			24	168	8766
Every minute		60	1440	10080	525960
Every second	60	3600	86400	604800	31557600

Possible frequencies of m

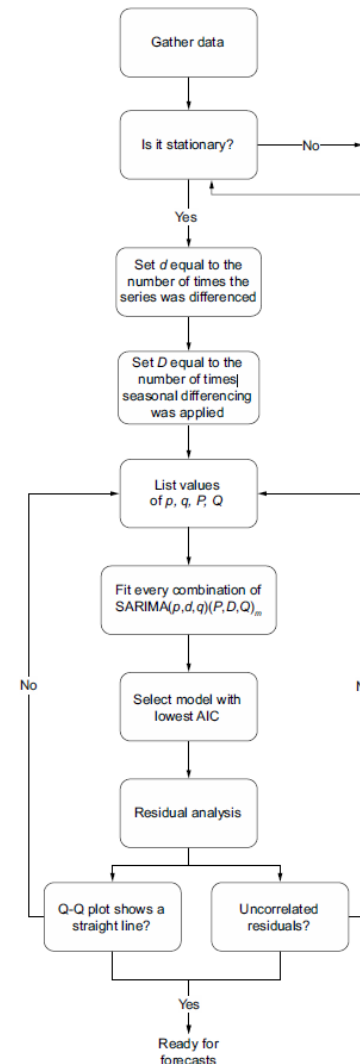


Seasonal AutoRegressive Integrated Moving Average (SARIMA)

- E.g.: $m = 12$, $P = 2$, $Q = 3$, $D = 1$
 - Including **two** past values of the series at lag that is multiple of $m \rightarrow y_{t-12}$ and y_{t-24} .
 - Including **three** past error terms at lag that is multiple of $m \rightarrow \epsilon_{t-12}, \epsilon_{t-24}, \epsilon_{t-36}$.
 - Seasonal difference will be applied: $y_t' = y_t - y_{t-12}$.



Seasonal AutoRegressive Integrated Moving Average (SARIMA)



General modeling procedure for SARIMA

Figure 8.11
procedure for



SARIMA - Seasonal Differencing

Is it important the **order** for D vs d?

→ Mathematically does not change the final result

d first → historical reasons

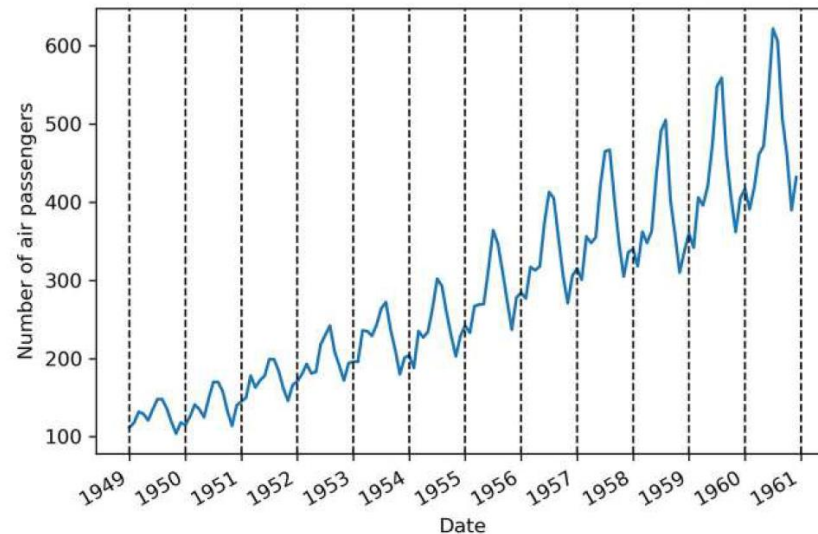
D first → remove most non-stationarity for cases of strong seasonality

(many series appear non-stationary because of seasonality, not trend)



Identifying seasonal patterns in a time series

- We should apply SARIMA on data that exhibits a seasonal pattern.
- How can we determine if there is a seasonal pattern?
 - Plotting the time series and observing if there are periodic patterns.

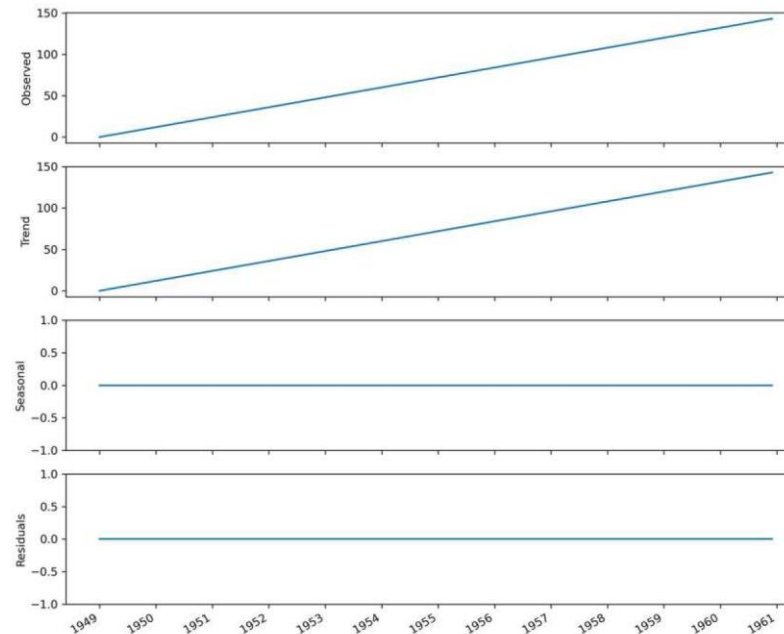
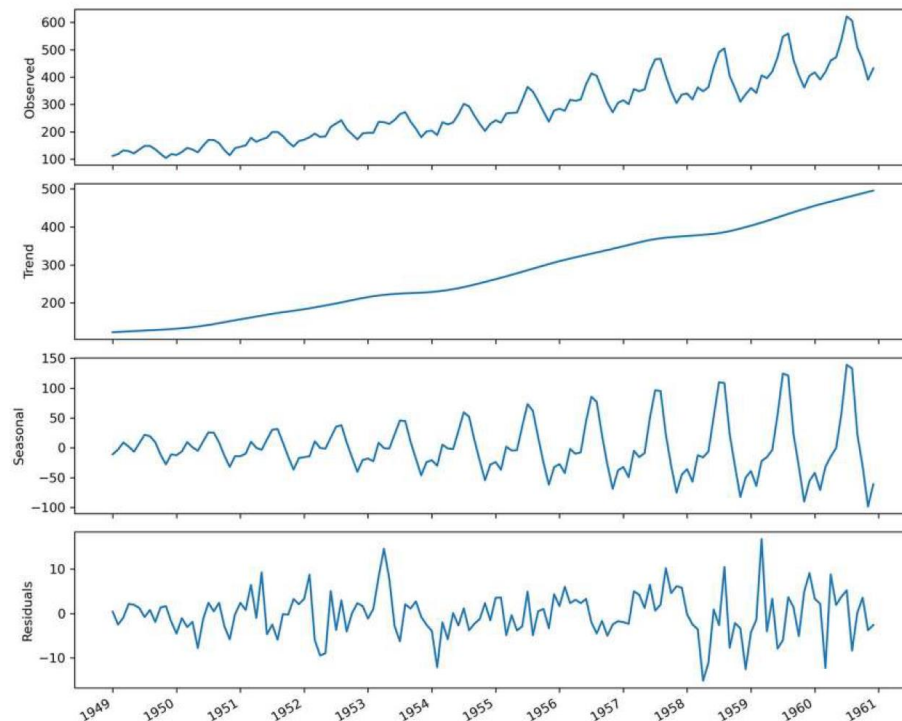


Seasonal pattern in the monthly number of air passenger. The peak occurs in the middle of each year: June, July, August... Similar at the end and beginning of each year.



Identifying seasonal patterns in a time series (II)

- How can we determine if there is a seasonal pattern?
 - Using time series decomposition: trend, seasonal and residual components.





Identifying seasonal patterns in a time series (II)

- How can we determine if there is a seasonal pattern?
 - **Canova-Hansen test** → Seasonal pattern is deterministic and stable hypothesis
 - **OCSB test** → seasonal unit root hypothesis



Seasonal AutoRegressive Integrated Moving Average (SARIMA)

- [Colab.](#)



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SARIMAX



Adding external variables

- So far, **all** the **models** we have been seeing to produce forecasts considered **only** the **time series itself** as data.
 - However, it is possible to add **external variables** or **exogenous variables**.
- SARIMAX: SARIMA + **Exogenous** variables.
 - **Endogenous variables**: the target variable we are trying to predict.
 - **Exogenous variables**: predictors, external or input variables

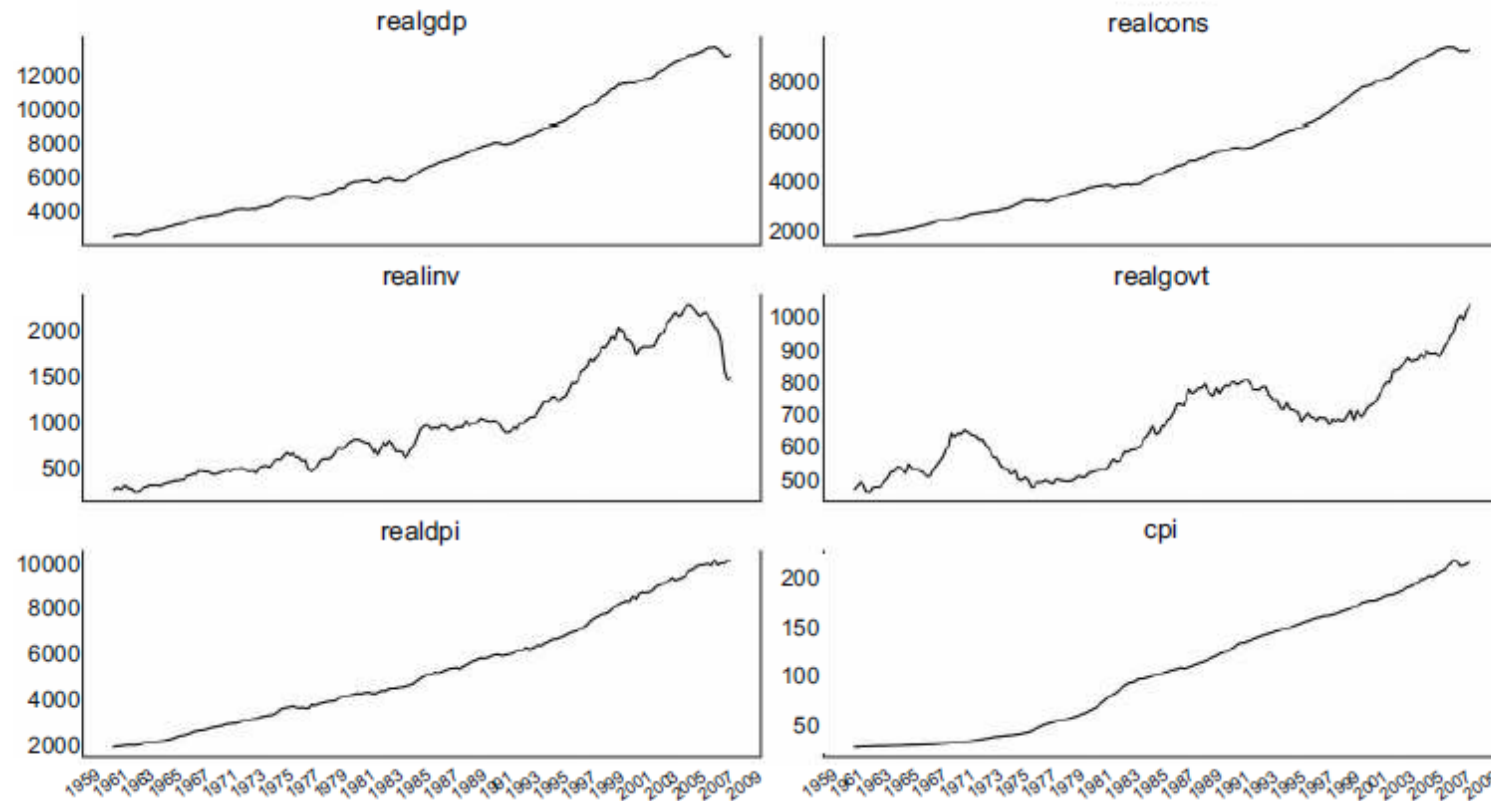


Adding external variables (II)

- Example:
 - Macroeconomic dataset from the United States.
 - Gross Domestic Product (**GDP**) every **quarter** from 1959 to 2009.
 - **Exogenous** data: consumption, government spending, investments and exports.
 - **Endogenous** data: the GDP.



Adding external variables (III)



Evolution of the real GDP and five exogenous variables
from 1959 to 2009.



SARIMAX

- Extends SARIMA model by adding exogenous variables.
 - SARIMA is a linear model: a combination of past values of the series and error terms.
 - With the exogenous variables, we are adding another linear combination.

$$y_t = SARIMA(p, d, q)(P, D, Q)_m + \sum_{i=1}^n \beta_i X_t^i$$

- We can encode categorical values too, but we need to convert them to numbers
- It can be considered as the most general function for forecasting time series.
 - We have been using it on all the Google Colabs so far :)



SARIMAX + External variables

- Caveat when forecasting multiple timesteps into the future.
- SARIMAX = SARIMA + Linear Combination of exogenous variables to predict one time step into the future.
 - What if... we wish to predict two timesteps into the future?
 - We must predict the exogenous variables too.



SARIMAX + External variables

- [Colab.](#)



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Other time series
forecast models



Libraries

Huge amount of resources for timeseries forecasting
[awesome-time-series](#)



Libraries – auto-ARIMA

- Simplest improvements, automatic SARIMAX:
[pmdarima](#)
- Based on Hyndman-Khandakar algorithm, brute-force search and statistical tests instead of qualitative results (such as ACF)
- D, Canova-Hansen test
- d, KPSS test
- AIC for p,q,P,Q
- Iterative method or grid search



Libraries – auto-ARIMA

- Iterative method (ARIMA version):
 - Test simplest 4 combinations: (2,d,2) (0,d,0), (1,d,0), (0,d,1)
 - Select the best (based on AIC)
 - Add more couple of variations (small changes ± 1) and retest AIC
 - Loop again until the all variations are worse than previous one
 - Doesn't mean we got the best possible parameters, local minima
- Grid Search:
 - More compute, but guarantees mathematical optimum



Libraries – basic ML

- We can use different lags of our time series to create features as dataframes and use simple classical ML models to predict the next timestep



Libraries – DL forecast

- Most used architectures:
 - GRUs & LSTMs (sequential data, memory based)
 - 1D-CNNs (moving window, less long term based)
 - Self-attention based (PatchTST, Informer, Autoformer)
 - Check entire timeline simultaneously checking the most related (“attention”)
 - Memory hungry and high complexity $O(N^2)$
 - Usually beats LSTM architectures
 - MLP based (N-BEATS, N-HiTS, TiDDE)
 - Surprisingly can beat transformers in some cases
 - Prone to overfitting without regularization techniques
 - As it is faster, can process a high amount of variables without much problem



Libraries – DL forecast

- Informer, 2020
 - Reduces complexity from $O(N^2)$ to $O(N \log N)$ by only selecting the most important connections
- Autoformer, 2021
 - Deep decomposition into trend and seasonal patterns for each layer
 - Replaces attention with an auto-correlation mechanism
- PatchTST, 2023
 - Group data into patches to feed it into transformers, reducing memory usage
 - If multiple metrics (covariate), it processes them independently initially



Libraries – DL forecast

- N-BEATS, 2019
 - Massive stack with residual links
 - Can be forced to be interpretable
 - Univariate instead of covariate
- N-HiTS, 2022
 - Multi-rate data sampling. Each block looks at the data with different frequencies (daily, weekly, etc.)
 - Originally univariate, adaptation for covariate
- TiDE, 2023
 - MLP Encoder-Decoder structure
 - Black box, no like previous ones
 - Very fast, highly scalable, multivariate forecasting



Libraries – DL forecast

- [Darts](#)
- [Nixtla](#)
- [PyTorch Forecasting](#)



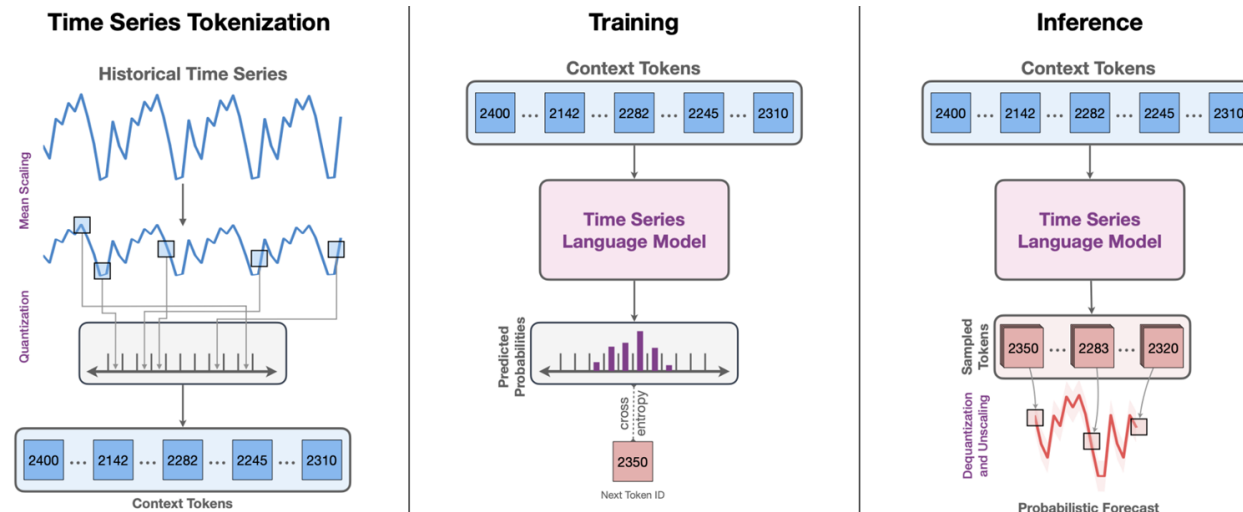
Libraries – Foundation Models

- Pretrained, general-purpose models (similar to LLMs)
 - Trained on all available time-series data on the internet + synthetic
 - Ability for zero-shot forecasts (without training for specific dataset)
 - Heavier than other models, but easy to use and overall good performance
- [Chronos-2](#), 2025, Amazon
 - Based on T5 architecture
 - [Github](#), [tutorial](#)
 - State of the art
 - Latest improvement allows covariate forecasting
 - Although heavier than others, only 120 million parameters, ~240MB of VRAM)



Libraries – Chronos

- Conversion of real-valued time series into a language-like format. Two steps:
 - **Scaling:** Each timeseries has its own magnitude/scale. Need to normalize so that it is generalizable. Scale series by absolute mean value
 - **Quantization:** This is where the **continuous scaled values** are made **discrete**. The entire potential range of the scaled time series values is **divided** into a **predefined number** of equally spaced **intervals**, known as **bins**.





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